*ms*2 – user manual

A Molecular Simulation Tool for Thermodynamic Properties

version 3.0

Kaiserslautern Paderborn, July 2017[a]

Main author: Simon Stephan

[a]The most up-tp-date version of the *ms*2 manual is available for download at *http://www.ms-2.de*.

Contents

1 Introduction

1.1 Program summary

*ms*2 has been successfully used in the Industrial Fluid Properties Simulation Challenge [fluidproperties 2010]. It is a molecular simulation program that aims at the calculation of thermodynamic properties that are needed for applications in the chemical industry, without requiring expert knowledge by the user. It was developed for rigid molecular models with a focus on condensed phases, covering mixtures containing an arbitrary number of small molecular species.

The simulation program *ms*2 is optimized for a fast execution on a broad range of computer architectures, spanning from single processor PCs over PC clusters and vector computers to high end parallel machines. It is a standalone Fortran90 code that does not require any libraries and does not have any other software prerequisites. *ms*2 is provided as a source code and it only needs to be compiled. The structure of *ms*2 is modular and object-oriented, except at the very core of energy and force calculations, where some structural compromises were made to achieve a better performance.

*ms*2 is aimed at homogeneous bulk systems in equilibrium and contains a molecular dynamics (MD) and Monte Carlo (MC) sampling. For vapour-liquid equilibrium (VLE) calculations, the coexisting phases are simulated independently and sequentially. Thus, it is usually sufficient to sample molecular systems containing on the order of 10^3 molecules to obtain statistically reliable results [Lyubartsev et al. 2000]. Applying standard cut-off radii, the intermolecular interactions are evaluated roughly up to the maximal distance, which is given by the edge length of the cubic simulation volume. Therefore, spatial decomposition schemes for parallelization are not rewarding. Instead, Plimpton's method [Plimpton 1993] was implemented for parallel execution of MD, which scales reasonably well up to 16 or 32 processors, depending on the particular architecture. For MC, an optimal parallelization scheme was introduced, taking advantage of the stochastic nature of such simulations. The computing resources are splitted into single runs each on one node, sampling the phase space independently. Both parallelization methods allow for short response times[1]. A typical molecular simulation of a VLE state point takes roughly six hours on a current workstation. The main expenses for determining thermodynamic data in silico are installation and maintenance of computing equipment and/or computing time, which is presently below US$ 0.35/CPU-hour [*The Economist* 2010].

The simulation program *ms*2 presented here was developed in a long-standing and ongoing cooperation of engineers and computer scientists. With the current version 3.0 of *ms*2, it is intended to facilitate the transfer of a state of the art and user-friendly molecular modeling and simulation package to academic institutions and the chemical industry. The release package consists of the simulation program *ms*2 itself and auxiliary feature tools for setup and analysis of simulation runs, including a simple 3D visualization tool to monitor the molecular trajectories. Furthermore, accurate force field models for more than 100 small molecules are provided.

1.2 Citation information & license

*ms*2 is a noncommercial software, which was published in Computer Physics Communications. The most current manuscript as the preceding ones along with their Supplementary Materials can be

[1] Under "response time" we understand the real time difference between submission and termination of a simulation run.

downloaded from the *ms*2 home page (*http://www.ms-2.de*). It is free of charge for scientific research and can be obtained after registration. Users need to acknowledge the use of *ms*2 by citing the following publication:

> S. Deublein, B. Eckl, J. Stoll, S. Lishchuk, G. Guevara-Carrion, C.W. Glass, T. Merker, M. Bernreuther, H. Hasse, J. Vrabec
>
> Computer Physics Communication **182** (2011) 2350
>
> doi:10.1016/j.cpc.2011.04.026

1.3 Support, development & maintenance

Supply the version and compiling details with a support request.

The newest version of the *ms*2 molecular simulation package is available at the *ms*2 homepage:

> *http://www.ms-2.de*

The source code of *ms*2 can be downloaded there as a tarball.

The following email address is the general contact point:

> *contact@ms-2.de*

*ms*2 users are encouraged to report bugs to the aforementioned email address.

2 Installation

2.1 Downloading the source code

The *ms*2 distribution is available on the webpage: http://www.ms-2.de/download. Users must either be a natural person or an academic institution represented by a natural person. The *ms*2 distribution consists of four major parts: the source code, java tools for pre- and postprocessing, force field models from the *Molecular Models of the Boltzmann-Zuse Society* database as input files for *ms*2 and some example input files for the calculation of different properties. The installation and use of the *ms*2 program is explained in detail in section 3, 4 to 5. The application of the java tools is shown in section 8. A detailed list of the force fields that comes with the *ms*2 distribution is in section 6.

2.2 Building *ms*2

*ms*2 is tested on Linux and Windows platforms, a short explanation how to install the *ms*2 on these two systems is given in the following.

2.2.1 Building *ms*2 on Linux platforms

The compilation can be carried out using the provided “Makefile” found in the “src” directory (e.g. with GNU make) and the compiler-related `*.mk`-files in the `makefiles`-subdirectory. The “Makefile” and the “`*.mk`”-file should be adopted to the local system requirements as described below. Alternatively (or additionally) variables can be set through the make command line options as compiler flags. An overview of all options can be found in table 2.1.
When compiling *ms*2, a variety of settings can be specified according to the users needs and optimise the performance. The following strings are recognized in the makefile: `TARGET`, `RELEASE`, `MPI`, `OMP`, `TRANS`, `HBOND`, `OSMOP`, `SP` and `PROG`. After settings these variable according to the users hardware and computational need, typing `make` in the command line from the “src” directory compiles *ms*2 and creates and executable in the current directory.

Settings in the `Makefile`
First of all a compiler has to be chosen. Through the definition of the `COMPILER` variable, the corresponding file with suffix “.mk” in the “makefiles” subdirectory will be included during the execution of `make`. These files contain specific settings for compilers and/or systems that *ms*2 is suitable for.

- Version:
  `TARGET = RELEASE` or `DEBUG`
  The `DEBUG` version is manly used by developers. Default is `RELEASE`.
- FORTRAN Compiler:
  `COMPILER = INTEL`, `GNU` or `PGI`
  The entry `GNU` found in the given Makefile corresponds to “makefiles/GNU.mk”, which contains settings for the widely used GNU compiler. There are other predefined settings available, like “INTEL” for the INTEL compiler. The default value is `GNU`, since that compiler is freeware.

- Parallelisation protocol:
  `MPI = 1` or `0` or
  `OMP = 1` or `0`
  These parameters have to be set to invoke the implemented parallelisation schemes.

The make command produces one executable file of *ms*2. To optimise the execution speed of the program for each application, the user should set the Feature Compiler Flags `TRANS`, `HBOND`, `OSMOP` and `SP` according the Simulation dependencies – especially if the user wants to run large numbers of simulations with one executable. We suggest only to set a `Compiler Variable = 1` if you want to use this certain feature, otherwise they will slow down your simulation speed.

- Transport properties:
  `TRANS = 1` or `0`
  To produce an executable that samples transport properties (like diffusion and conductivity) `TRANS = 1` has to be set. A value of `0` means that the routines for the calculation of transport properties are not compiled. The default value is `0`, since the program execution for the sampling of the default properties is faster in that case.

- Hydrogen-bonding evaluation:
  `HBOND = 1` or `0`
  If you want to have a look at the H-Bonds in your system, turn this on (set variable `1`). It enables you to specify a geometric criterion for H-bonds in your input files. *ms*2 plots then the pursuant H-bonding statistics. A Value of `0` means, that the routines for the calculation of H-bonds are not to be compiled. The default value is `0`, since the program execution for the sampling of the default properties is faster in that case.

- Osmotic pressure:
  `OSMPO = 2`, `1` or `0`
  This flag enables the osmotic pressure method. The value `2` enables the pressure profile along the density profile and osmotic pressure results. A Value of `0` means, that the routines for use of the osmotic pressure method are not to be compiled. The default value is `0`.

- Single precision:
  `SP = 1` or `0`
  To speed up your simulation you can choose single precision (set value `1`) instead of the default double precision (set value `0`). Users are advised to use `SP = 1` with care.

- Name of the executable file:
  `PROG = String`
  The user can set the value to adapt the name of the executable file that is produced by the make command. According to the values of `TRANS`, `HBOND`, `OSMOP`, `SP`, `COMPILER` and `TARGET` the executable will have an addition in its name respectively.

Variable	Description	Values	Description
Compiler settings:			
`COMPILER`	Compiler presets	`GNU`	GNU compiler
		`INTEL`	INTEL compiler
		`PGI`	PGI compiler
		see makefiles subdirectory	
`TARGET`	see above	`RELEASE`	for production simulations
		`DEBUG`	for code developing
Parallelization setting:			
`MPI`	MPI	`0`	off
		`1`	on
`OMP`	OpenMP	`0`	off
		`1`	on
Features to be accessible:			
`SP`	Single Precision	`0`	off
		`1`	on
`TRANS`	Transport	`0`	off
		`1`	on
`OSMOP`	Osmotic pressure	`0`	off
		`1`	on
`HBOND`	H-Bonding statistics	`0`	on
		`1`	on
`PROG`	Name of executable	String	can be set by user

Table 2.1: Makefile definitions

Compiler-related `*.mk` files

For each supported compiler, the subdirectory `makefiles` in `src` contains one `*.mk` file, e.g. `INTEL.mk`. The first three lines define the used FORTRAN90 comilers. In rare system depending cases, the optimisation degree (the integer after the -O for the `F90FLAGS_RELEASE` has to be decreased to 2 or 1). It is suggested to use the slightly faster `mpiifort`-Compiler instead of the `mpiF90` in case you are using the `INTEL.mk`.

make command line options

Instead of changing the Makefile, the definitions can also be set directly in the command line, that will overwrite the settings in the makefile. For example:

```
make TARGET=RELEASE COMPILER=GNU
```

Compiler Flags Depending on compiler and preprocessor variables set in the makefile, the following compiler flags are used in the compilation/build process:

OMPFlags: -mp/openmp/fopenmp

CPPFlags: -DARCH=2 –DFORTRAN=90 (replaces compiler flags for ARCH and FORTRAN)

Releaseflags: -O3 -vec/qopt-report=3/5

Debugflags: -O0 -vec/qopt-report=0 -g

GNU: -ffree-line-length-none -mtune=generic

INTEL: -fpp –r8

PGI: -C

If the user is unsure what compiler flags to set, it is suggested to leave the make file blank.

2.2.2 Building on Windows

*ms*2 can also be executed on windows systems. The source code has to be downloaded and compiled in a similar way as for Linux platforms. It is suggested to use a Linux-like shell, for example *cygwin*. It provides a terminal window that runs a handsome Linux shell and comes with the *GNU gfortran* compiler[1]. This is very useful, since *ms*2 is a command line-based program whose utilisation is in our experience best in combination with the standard Linux commands, such as `sed`, `awk`, `grep`, `find` etc.. Additionally, the preparation of input files and analysis of the output files of *ms*2 can be done by shell scripts. The installation of *ms*2 in *cygwin* follows the installation instructions on Linux platforms discussed above. Of course, *ms*2 can also be compiled using standard windows tools.

[1] during the installation of cygwin, all the FORTRAN depending packages have to be tagged.

3 Running *ms*2

*ms*2 was designed to be an easily applicable simulation program. Therefore, the input files are restricted to one file for the definition of simulation scenario and one file for each of the molecular species that are used in the simulation. The output files contain structured information on the simulation. All calculated thermodynamic properties are summarized in one output file, which is straightforwardly readable and self-explaining. For a more detailed evaluation of the simulation, the instantaneous and running averages of the most important thermodynamic properties are written to other files. In total, the current status of the simulation and many more details are written to six files and can be accessed during execution. The structure of the input files as well as the output files is shown in Figure 3.1.

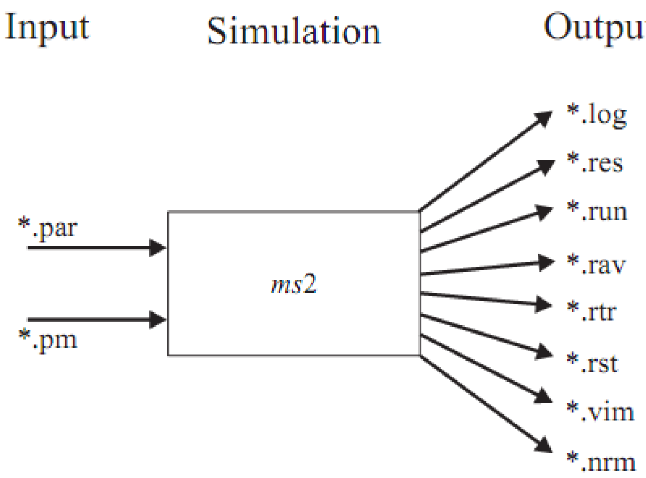


Figure 3.1: File structure needed and generated by *ms*2.

3.1 Input structure

*ms*2 expects two input files: the `*.par` and the `*.pm` file. The `*.par` file specifies the simulation parameters and the `.pm` file the molecular potential model. The `*.par` file is the main file steering the actual simulation. It contains all input variables for the simulation process, such as simulation type, ensemble, number of equilibration and production steps, time step length etc. Also the potential models need to be declared here. Each `*.pm` file specifies exactly one chemical substance. To calculate mixtures, several `*.pm` files have to be specified in the `*.par` file.

Each line in both input file types sets one parameter, is empty or contains a comment. The comment symbol # and everything after it up to the end of the line is ignored by *ms*2 in the same way as empty lines are. It is suggested to use only blanks instead of tabs, since the latter may cause trouble using input files generated on other platforms.

A line setting a parameter in either `*.par` or `*.pm` file always consists of a keyword `XXX`, the = symbol and the parameter value YYY, e. g.
`NVTSteps = 100000`
Leading spaces, trailing spaces and spaces surrounding = symbol are ignored. Parameter values can be

strings in some case or numeric values. Parameter names and values that *ms*2 expects for the parameter file and potential model files are described in sections 4. All parameters must be present in the input files in the certain order. The program reads the input file sequentially. Unidentified parameters are ignored. Missing or misplaced parameters lead to the termination of the program with the corresponding error message.

3.1.1 Parameter file

Below is an exemplary *`.par` file for a MD simulation of pure oxygen in the NVT ensemble. *`.par` files are organised in three sections: the first section sets the general simulation parameters, such as timestep, MD or MC, unit systems etc. (lines 3 to 20 in the oxygen example), the second section sets the thermodynamic quantities to define the ensemble, such as temperature, density, number of components etc. (lines 21 to 26). The third section contains the potential models that shall be simulated and indicate their mole fraction (lines 28 to 31). The value for the keyword `PotModel` needs to be set according to the name of the potential model file that shall be used, including the tailing *`.pm`. Each *`.par` file ends with the cutoff that shall be applied throughout the simulation.

This example runs a MD simulation of pure oxygen at a constant temperature of 40 K with an equilibration of 100.000 timesteps and a production period of 1.000.000 timesteps. The simulation contains 1372 molecules and uses a cutoff distance of 15 Å.

*.par file for O_2 simulation

```
1  # ms2 parameter file
2
3  Units          =       SI
4  LengthUnit     =       1.0
5  EnergyUnit     =     100.0
6  MassUnit       =      10.0
7  Simulation     =       MD
8  Integrator     =       Gear
9  TimeStep       =       0.001
10 Ensemble       =       NVT
11 MCORSteps      =       0
12 NVTSteps       =  100000
13 RunSteps       = 1000000
14 ResultFreq     =    1000
15 ErrorsFreq     =    5000
16 VisualFreq     =       0
17 CutoffMode     =       COM
18 LongRange      =       ReactionField
19 NEnsembles     =       1
20
21 # Ensemble 1
22 Temperature    =      60.0
23 Density        =      40.664
24 NParticles     =    1372
25 NComponents    =       1
26
27 # substances
28 PotModel       =       O2.pm
29 MoleFract      =       1.0
```

```
ChemPotMethod =       none

Cutoff        =      15.00
```

Table 4.1 lists all input parameters and options to be specified in the *.`par` file in section 4. The implemented methods behind these keywords are documented in section 5. Creating a *.`par` file is facilitated by the *ms*2 feature program *ms2par*. Its use is described in detail in section 8.1.

3.1.2 Potential model file

A *.`pm` file contains the molecular model, also called force field, for a given substance. It contains the relative positions and parameters of all interaction sites. The potential model file for methanol is shown below. Methanol was modeled by two LJ sites and three point charges [Schnabel et al. 2007a].

All positions and distances in the *.`pm` file are given in Å, the Lennard-Jones interaction parameters σ and ϵ/k_B are given in Å and K, respectively, while the mass is given in atomic units (u= 1.6605 $\times 10^{-27}$ kg). The magnitudes of the charges are specified in electronic charges (e= 1.602×10^{-19} C), while dipole moments and quadrupole moments are given in Debye (D = 3.33564×10^{-30} Cm) and Buckingham (B = $3.33564 \times 10^{-40}\mathrm{Cm}^2$), respectively. The orientations of the dipole and quadrupole are represented by spherical coordinates, where the azimuthal angle ϕ specifies the angle to the positive x axis and the polar angle θ defines the angle to the positive z axis. Both angles are specified in degrees. Molecular models can be orientated arbitrarily in the *.`pm` file. All site positions are automatically transformed into a principal axes coordinate system at the beginning of each simulation run by *ms*2. The normalized site positions are written to a *.`nrm` output file for each component.

`*.pm file for methanol`

```
# ms2 potential model file

NSiteTypes   =  2

SiteType     =  LJ126
NSites       =  2

# CH3
x            =  7.660331e-01
y            =  1.338147e-02
z            =  0.0
sigma        =  3.7543
epsilon      =  120.592
mass         =  15.035

# O
x            =  -6.564695e-01
y            =  -6.389332e-02
z            =  0.0
sigma        =  3.030
epsilon      =  87.879
```

```
mass          =  15.999

SiteType      =  Charge
NSites        =  3

# H[e(1)]
x             =  -1.004989e+00
y             =  8.145993e-01
z             =  0.0
charge        =  +0.43128
mass          =  1.008

# e(2)
x             =  -6.564695e-01
y             =  -6.389332e-02
z             =  0.0
charge        =  -0.67874
mass          =  0.0

# e(1)
x             =  7.660331e-01
y             =  1.338147e-02
z             =  0.0
charge        =  +0.24746
mass          =  0.0

NRotAxes    =  auto
```

All input parameters and options to be specified in the *.pm file are given in table 3.1. A *.pm file has a specific structure, which is described in the following. The parameter `NSiteTypes` gives the number of different potential types (not the number of sites). Different potential types are 12-6 Lennard-Jones sites, point charges, point dipoles and point quadrupoles. Then the descriptions for each potential type follow sequentially. Each of them contains the number of sites of corresponding type `NSites`, followed by the definition of each individual site. Each site requires x, y and z coordinates and the mass in addition to its interaction parameters, depending on the site type. Lennard-Jones sites require one size and one energy parameter, point charges only the magnitude of the charge, while point dipoles and point quadrupoles require magnitude and orientation. At the end of a *.pm file, the parameter `NRotAxes` is specified. Parameters `InertMomX`, `InertMomY` and `InertMomZ` are read if the moments of inertia are to be specified manually, set according to the `NRotAxes` value.

keyword	value	explanation	recommended value
NSiteTypes	*int*	number of site types to appear in *.pm file	
SiteType	LJ126 Charge Dipole Quadrupole	type of interaction site	must be specified
NSites	*int*	number of sites for the corresponding interaction site type	must be specified
x	*double*	spatial coordinate of a site in Å	must be specified for each site
y	*double*	spatial coordinate of a site in Å	must be specified for each site
z	*double*	spatial coordinate of a site in Å	must be specified for each site
theta	*double*	angle θ between the dipole or quadrupole and the positive z axis in degree	must be specified for dipole or quadrupole site
phi	*double*	angle ϕ between the dipole or quadrupole and the positive x axis in degree	must be specified for dipole or quadrupole site
sigma	*double*	Lennard-Jones radius σ in Å	must be specified for Lennard-Jones site
epsilon	*double*	Lennard-Jones energy ε/k_B in K	must be specified for Lennard-Jones site
charge	*double*	magnitude of point charge site	must be specified for charge site
dipole	*double*	dipole moment for dipole site	must be specified for dipolar site
quadrupole	*double*	quadrupole moment for quadrupole site	must be specified for quadrupolar site
mass	*double*	mass in atomic units	must be specified for each site
shielding	*double*	shielding radius in Å for MC steps	can be specified to get better convergence in MC run
NRotAxes	auto 0 2 3	number of rotation axes of the molecule	auto
TotalMass	*double*	mass of molecule in atomic units	optional
InertMomX	*double*	component of inertia moment in atomic mass unit multiplied by Å^2	if NRotAxes is 2 or 3
InertMomY	*double*	component of inertia moment in atomic mass unit multiplied by Å^2	if NRotAxes is 2 or 3
InertMomZ	*double*	component of inertia moment in atomic mass unit multiplied by Å^2	if NRotAxes is 3

Table 3.1: Parameters and options specified in a *.pm file.

3.2 Output structure

*ms*2 yields different output files depending on the specifications made in the *.`par` file:

- *.`log` file - stores a complete summary of all execution steps taken by *ms*2.
- *.`res` file - contains the results of the simulation in an aggregated form. The data are written to file in reduced quantities as well as in SI units, along with the statistical uncertainties of the calculated properties. The *.res file is created during simulation and updated every specified number of time steps or loops.
- *.`run` file - contains the calculated properties of the simulation for a specified time step or loop interval. The file is in tabular form, where the data are given in reduced units. The file is subsequently updated according to the user specification, which is set in the *.par file.
- *.`rav` file - contains the block averages of the calculated properties. The file is in tabular form, where the data are given in reduced units. The file is updated according to the user specification, which is set in the *.par file.
- *.`rst` file - is the restart file of the simulation. It contains all molecular positions, velocities, orientations, forces, torques and block averages for the thermodynamic properties. It is written once at the end of a simulation or immediately after having received a termination signal of the operating system. The *.rst file allows for a stepwise execution of the simulation, necessary e.g. in case of an early interruption of the simulation, time limits on a queuing system or unexpected halts.
- *.`vim` file - is the trajectory visualization file. It contains the positions and orientations of all molecules in an aggregated ASCII format. The configurations are written to file after a user-specified interval of time steps or loops. The *.vim file is readable by the visualization tool *ms*2*molecules*, which is also part of the simulation package.
- *.`nrm` file - stores the normalized coordinates of a potential model after a principal axes transformation.
- *.`rtr` file - stores the final value of the autocorrelation functions and their integrals. The number of output lines is equal to Corrlength divided by StepsCorrfun. The number of output lines has to be defined in the *.par file.

4 Keywords for parameter (.par) files

Table 4.1 gives an overview over the keywords (first row) that can be specified in the *`.par` file and its expected values (second row). The third row gives a short explanation of the keyword, while the fourth row gives some recommendations.

keyword	value	explanation	recommended value
Simulation specific:			
`WallTime`			`1420`
`TimeLimit`			`120`
`Units`	`SI`	physical properties in the *.par file are given in SI units	`SI`
	`reduced`	physical properties in the *.par file are given in reduced units with respect to the reference values of length σ_{ref}, energy ϵ_{ref} and mass m_{ref}	
`LengthUnit`		reference length σ_{ref} in Å	`3.0`
`EnergyUnit`		reference energy ϵ_{ref}/k_B in K	`100.0`
`MassUnit`		reference mass m_{ref} in atomic units $u = 1.6605 \times 10^{-27}$ kg	100.0
`Simulation`	`MD`	molecular dynamics simulation	
	`MC`	Monte-Carlo simulation	
`Integrator`	`Gear`	gear predictor-corrector integrator (MD only)	`Gear`
	`Leapfrog`	Leapfrog integrator (MD only)	
TimeStep		time step of one MD step in fs (MD only)	$\sim$1
`Acceptance`		acceptance rate for MC moves (MC only)	`0.5`
`Ensemble`	`NVT`	canonical ensemble	
	`NVE`	micro-canonical ensemble	
	`NPT`	isobaric-isothermal ensemble	
	GE	Grand equilibrium method (pseudo-μVT)	
	`NpH`	isoenthalpic–isobaric ensemble	
`MCORSteps`		MC relaxation loops for pre-equilibration	`100`
`NVTSteps`		number of equilibration time steps (MD) or loops (MC) in the NVT ensemble	`20000`
`NPTSteps`		number of equilibration time steps (MD) or loops (MC) in the NpT ensemble (optional)	`50000`
`RunSteps`		number of production time steps (MD) or loops (MC)	`300000`
`ResultFreq`		size of block averages in time steps or loops	`100`
`ErrorFreq`		frequency of writing the *.res file in time steps or loops	`5000`
`VisualFreq`		frequency of saving trajectories in the *.vim file	`5000`
`RDFFreq`			`1000`
`CutoffMode`	`COM`	center of mass cut-off	`COM`
	`Site`	site-site cut-off	
`NEnsemble`		number of ensembles in the simulation	1

Continued on next page

keyword	value	explanation	recommended value
mpiEnsembleGroups			1
CorrfunMode	yes	calculation of autocorrelation functions enabled	
	no	calculation of autocorrelation functions disabled	
Temperature		specified temperature	
Pressure		specified pressure	
Density		specified density	
PistonMass		piston mass for simulations at constant pressure	
NParticles		total number of molecules	864 - 4000
NumShells		number of shells for the calculation of RDF	300
OptPressure			
CommonEqui			
NComponents			
Liqdensity		simulation result density	
VarLiqDensity		statistical uncertainty of density	
LiqEnthaly		simulation result residual enthalpy	
VarLiqEnthaly		statistical uncertainty of residual enthalpy	
LiqBetaT		simulation result isothermal compressibility	
VarLiqBetaT		statistical uncertainty of isothermal compressibility	
LiqdHdP		simulation result $(dh/dp)_T$	
VardHdP		statistical uncertainty of $(dh/dp)_T$	
NComponents		number of components	
Corrlength		length of the autocorrelation in time steps	
SpanCorrFun		time steps separating subsequent autocorrelation functions	
ViewCorrFun		output frequency of the full autocorrelation functions into the *.rtr file	
ResultFreqCF		output frequency of transport properties into the *.res file	
PotModel		potential model *.pm file of a component	
MolarFract		molar fraction of a component	
ChemPotMethod	none	no calculation of the chemical potential for this component	
	Widom	calculation of the chemical potential for this component using Widoms's test molecule method	
	ThermoInt	calculation of the chemical potential for this component using thermodynamic integration	
NTest		number of test molecules for Widom's test molecule method	2000
WeightFactors	Guess	for gradual insertion: use user defined initial values for the weight factors with optimization of these factors during simulation	Guess
	OptSet	for gradual insertion: values for the weight factors without adjustment during simulation	

Continued on next page

keyword	value	explanation	recommended value
eta		binary size interaction parameter η – combination rule	1
xi		binary energy interaction parameter ξ – combination rule	1
NHBondCriteria			2
Cutoff		cut-off radius for center of mass cut-off	
CutoffLJ		cut-off radius for LJ interactions (site-site cut-off)	
CutoffDD		cut-off radius for dipole-dipole interactions (site-site cut-off)	
CutoffDQ		cut-off radius for dipole-quadrupole interactions (site-site cut-off)	
CutoffQQ		cut-off radius for quadrupole-quadrupole interactions (site-site cut-off)	
Epsilon		reaction field parameter	1.0E+10

Table 4.1: Parameters and options specified in the *.par file.

5 Methods

5.1 Basic simulation techniques

The basics of the molecular simulation techniques, e. g. molecular dynamics (MD) and Monte Carlo (MC), employed in *ms*2 are well described in the literature [Allen 2002; Frenkel 2002; Rappaport 2004] and are thus not repeated here. All non-standard methods that are implemented in *ms*2 are introduced in section 5. The following paragraph briefly describes the most significant assumptions and techniques employed in *ms*2.

Simulations with *ms*2 are performed in a quasi-infinite Cartesian space, using periodic boundary conditions [Born et al. 1912] and the minimum image convention [Metropolis et al. 1953]. The molecular interactions are considered to be pairwise additive, like in most other simulation packages. Including three- and many-body interactions leads to an increase in computational effort while the benefits are under discussion. As usual, the computational effort in *ms*2 is reduced by the introduction of a cut-off radius, up to which the intermolecular interactions are evaluated explicitly. The contributions of interactions with molecules beyond the cut-off radius are accounted for by correction schemes. For the electrostatic interactions, the reaction field and Ewald summation method is included, while for the dispersive and repulsive interactions, a homogeneous fluid beyond the cut-off radius is assumed. The long range contributions are added to all thermodynamic data that are calculated in *ms*2.

Many thermodynamic properties calculated with *ms*2 are residual quantities, indicated by the superscript "res". To compare them to data that are measured on the macroscopic level, the solely temperature dependent contributions of the ideal gas have to be added. These ideal properties are accessible e. g. by quantum chemical methods and can often be found in databases [Rowley 2003].

The simulation program *ms*2 internally uses reduced quantities for its calculations. All quantities are reduced by a reference length σ_R, a reference energy ϵ_R and a reference mass m_R, respectively. These reference values are input variables and may, in principle, be chosen arbitrarily. However, it is recommended to use a reference length σ_R in the order of 3 Å, a reference energy $\epsilon_\mathrm{R}/k_\mathrm{B}$ in the order of 100 K and a reference mass m_R in the order of 50,atomic units. The reduction scheme for the most important physical quantities is listed in table 5.1. From these properties, the reduced form of all other quantities can be derived. An exception in the reduction scheme is the chemical potential, which is normalized in *ms*2 by $k_\mathrm{B}T$ instead of ϵ_R.

quantities	reduced form
length	$l^* = \frac{l}{\sigma_\mathrm{R}}$
energy	$u^* = \frac{u}{\epsilon_\mathrm{R}}$
mass	$m^* = \frac{m}{m_\mathrm{R}}$
time	$t^* = \frac{t}{\sigma_\mathrm{R}}\sqrt{\frac{\epsilon_\mathrm{R}}{m_\mathrm{R}}}$
piston mass	$Q_\mathrm{p}^* = \frac{Q_\mathrm{p}\sigma_\mathrm{R}^4}{m_\mathrm{R}}$
temperature	$T^* = \frac{Tk_\mathrm{B}}{\epsilon_\mathrm{R}}$

Continued on next page

quantities	reduced form	
pressure	p^*	$= \frac{p\sigma_\mathrm{R}^3}{\epsilon_\mathrm{R}}$
density	ρ^*	$= \rho\sigma_\mathrm{R}^3 N_\mathrm{A}$
volume	V^*	$= \frac{V}{\sigma_\mathrm{R}^3}$
chemical potential	$\tilde{\mu}$	$= \frac{\mu}{k_\mathrm{B}T}$
point charge	q^*	$= \frac{1}{\sqrt{4\pi\epsilon_0}} \frac{q}{\sqrt{\epsilon_\mathrm{R}\sigma_\mathrm{R}}}$
dipole moment	μ^*	$= \frac{1}{\sqrt{4\pi\epsilon_0}} \frac{\mu}{\sqrt{\epsilon_\mathrm{R}\sigma_\mathrm{R}^3}}$
quadrupole moment	Q^*	$= \frac{1}{\sqrt{4\pi\epsilon_0}} \frac{Q}{\sqrt{\epsilon_\mathrm{R}\sigma_\mathrm{R}^5}}$
diffusion coefficient	D^*	$= \frac{D}{\sigma\sqrt{m_\mathrm{R}/\epsilon_\mathrm{R}}}$
viscosity	η^*	$= \frac{\eta\sigma_\mathrm{R}^2}{\sqrt{m_\mathrm{R}\epsilon_\mathrm{R}}}$
thermal conductivity	λ^*	$= \frac{\lambda\sigma_\mathrm{R}^2}{k_\mathrm{B}\sqrt{m_\mathrm{R}/\epsilon_\mathrm{R}}}$

Table 5.1: Important physical quantities in their reduced form. Note that ϵ_0 indicates the permittivity of the vacuum $\epsilon_0 = 8.854187 \times 10^{-12}\mathrm{A}^2\mathrm{s}^4\mathrm{kg}^{-1}\mathrm{m}^{-3}$.

5.2 Molecular dynamics

Molecular dynamics (MD) sample the physical movements of particles (atoms or molecules) of a given system. The path of the particles they follow through space as a function of time is determined from the forces acting between particles by numerically solving the Newton's equations of motion. These forces are described by atomistic force fields, i. e. the potential models of the substances. Besides offering insight into molecular motion and providing direct structural information on an atomic scale, MD allows the calculation of macroscopic thermodynamic properties directly from statistics and the acting forces and trajectories in the system. In other words: macroscopic properties are obtained from microscopic information. As MD follows the time evolution of the system, the outcome of a simulation is practically the time dependence of the calculated properties.
Transport properties can be obtained by *ms*2 only via molecular dynamics with the Green-Kubo formalism but not from Monte-Carlo sampling.

5.2.1 Implemented thermostat and barostat

The thermostat implemented in *ms*2 is the 'velocity scaling'. Here, the velocities are scaled such that the actual kinetic energy matches the specified temperature. The scaling is applied equally over all molecular degrees of freedom. The pressure is kept constant using Andersen's barostat in MD.

5.2.2 Integrators

In *ms*2, two integrators are implemented to solve Newton's equations of motion during the MD simulation: Leapfrog and Gear predictor-corrector. These integrators are well known and described in literature [Allen 2002]. The leapfrog integrator is a second order integrator that requires little computational effort, while being robust for many applications. The error of the method scales quadratically with the length of the chosen time step. The Gear predictor-corrector integrator is implemented with fifth order and is more accurate for small time steps compared to Leapfrog integration [Allen 2002]. The computational demand for both integration schemes is similar as implemented in $ms2$. The Gear integration, though being of higher order, is only 0.3% slower than the Leapfrog algorithm for the total calculation of one MD time step with a molecular model composed of three LJ sites, having three rotational degrees of freedom.

5.3 Monte Carlo

The Monte Carlo (MC) method is the application of statistical mechanics to describe molecular systems. Problems that can be simulated with MC can also be modeled by MD. The difference is that the MC approach relies on statistics rather than trying to reproduce the deterministic trajectory of particles through time. For MD the solution of Newton's equations of motion is the trajectory of every particle in the system, the trajectories can be interpreted as a sequence of states (configurations). The main idea behind MC is to create a set of configurations with Markov chain property such that this chain contains only relevant and physically meaningful configurations. In the standard MC simulation procedure, configurations of a given ensemble are generated in a stochastic way by performing trial changes in the system and then accepting them with appropriate criteria [Allen 2002].

Translation and rotation:
For MC, independent on the choice of ensemble, the generation of trial configurations always includes at least molecular translational and rotational motions. Each new configuration generated by these moves is either rejected or accepted, depending on the evaluation of its energy using a specific acceptance criterion. For a MC simulation that samples only translational and rotational molecular motions, one simulation loop in $ms2$ is defined as the total number of molecular degrees of freedom in the system divided by 3. In other words, during one step N randomly selected molecules are both translated and rotated assuming the number of degrees of freedom for each molecules is 6, where N is the number of molecules in the system.

Resize:
In an MC simulation, where the volume of the system does not remain constant, e. g. in a isobaric simulation run, the creation of trial configurations also includes, next to translational and rotational molecular motions, the adjustment of the simulation volume. This is done by rescaling every intermolecular distance such that the relative distances between molecules along with their spatial orientations remain unchanged. In $ms2$, during one simulation loop exactly one volume change attempt is made.

5.4 Interaction types

Currently four types of interaction sites are implemented: standard Lennard-Jones 12-6 sites, point charges, point dipoles and point quadrupoles. The first one models repulsion and dispersion, while charges, dipoles and quadrupoles model the electrostatic interactions of molecules, e. g. their polarity. A dipole is similar to two point charges, but leads to with less computational expense. In analogy, a point quadrupole is similar to a linear configuration of three point charges. Also the interaction of unlike sites is briefly described.

5.4.1 Dispersive and repulsive interactions

In $ms2$, the dispersive and repulsive interactions between molecules are reduced to pairwise interactions u_{ij}^{LJ} of the different molecule sites i and j, which are modeled by the 12-6 Lennard-Jones potential [Lennard-Jones 1931]:

$$u_{ij}^{\mathrm{LJ}} = 4\varepsilon \left(\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^{6} \right) . \tag{5.1}$$

Here, the site-site distance between two interacting LJ sites i and j is denoted by r_{ij}, while σ and ε are the LJ size and energy parameters, respectively. The LJ potential is widely used and allows for a fast computation of these basic interactions. It has only two parameters, which facilitates the parametrization of molecular models.

5.4.2 Electrostatic interactions

$ms2$ considers electrostatic interactions between electro-neutral molecules. Three different electrostatic site type models are available: point charge, point dipole and linear point quadrupole. The two higher order electrostatic interaction sites integrate characteristic arrangements of several single partial charges on a molecule. The simulative advantage of higher order polarities is faster execution and a better description of the electrostatic potential of the molecule for a given number of electrostatic sites [Stone 2008]. Combining two partial charges to one point dipole reduces the computational effort with $ms2$ by about 30%, combining three point charges to one point quadrupole reduces the computational demand by even 60%. An additional advantage is the simplification of the molecular model, which reduces the number of molecular model parameters. In the following, the implemented electrostatic interactions between sites of the same type and between unlike site types are briefly described.

Point charge
Point charges are first order electrostatic interaction sites. The electrostatic interaction between two point charges q_i and q_j is simply given by *Coulomb's law* [Rappaport 2004; Frenkel 2002].

$$u_{ij}^{qq}\left(r_{ij}, q_i, q_j\right) = \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}} . \tag{5.2}$$

This interaction decays with the inverse of r_{ij} and is therefore significant up to very large distances. For computational efficiency, the Coulombic contributions to the potential energy are explicitly evaluated up to a specified cut-off radius r_c. The long range interactions between charges beyond r_c are corrected for by the reaction field method or Ewald summation.

Point dipole
A point dipole describes the electrostatic field of two point charges with equal magnitude, but opposite sign at a mutual distance $a \to 0$. Its moment μ is defined by $\mu = qa$. The electrostatic interaction between two point dipoles with the moments μ_i and μ_j at a distance r_{ij} is given by [Allen 2002; Gray et al. 1984]:

$$u_{ij}^{DD}\left(r_{ij}, \theta_i, \theta_j, \phi_{ij}, \mu_i, \mu_j\right) = \frac{1}{4\pi\epsilon_0} \frac{\mu_i \mu_j}{r_{ij}^3} \left(\sin\theta_i \sin\theta_j \cos\phi_{ij} - 2\cos\theta_i \cos\theta_j\right) , \tag{5.3}$$

with θ_i being the angle between the dipole direction and the distance vector of the two interacting dipoles and ϕ_{ij} being the azimuthal angle of the two dipole directions, cf. Figure 5.1. In $ms2$, the interaction between two dipoles is explicitly evaluated up to the specified cut-off radius r_c. The long range contributions are considered in $ms2$ by the reaction field method or Ewald summation Allen 2002.

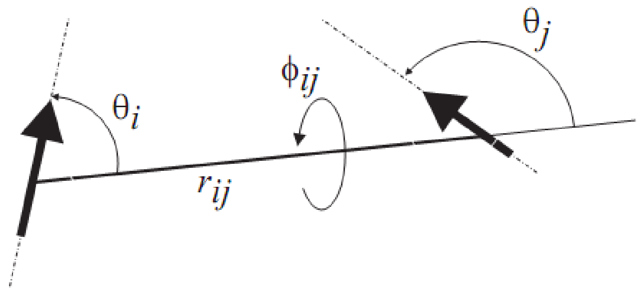


Figure 5.1: Schematic of the angles between two point dipoles i and j indicated by the arrows, which are situated on different molecules at a distance r_{ij}.

Linear point quadrupole

A linear point quadrupole describes the electrostatic field induced either by two collinear point dipoles with the same moment, but opposite orientation at a distance $a \to 0$, or by four collinear point charges with alternating sign $(q, -q, -q, q)$. The resulting quadrupole moment Q is defined by $Q = 2aq$. The electrostatic interaction between two linear point quadrupoles with the moments Q_i and Q_j at a distance r_{ij} is given by [Gray et al. 1984; Hirschfelder et al. 1954]

$$\begin{aligned} u_{ij}^{QQ}\left(r_{ij}, \theta_i, \theta_j, \phi_{ij}, Q_i, Q_j\right) \quad = \quad & \frac{1}{4\pi\epsilon_0}\frac{3}{4}\frac{Q_iQ_j}{r_{ij}^5} \\ & [1 - 5((\cos\theta_i)^2 + (\cos\theta_j)^2) - 15\,(\cos\theta_i)^2\,(\cos\theta_j)^2 + \\ & 2\,(\sin\theta_i \sin\theta_j \cos\phi_{ij} - 4\cos\theta_i\cos\theta_j)^2]\ , \end{aligned} \tag{5.4}$$

where the angles θ_i, θ_j and ϕ_{ij} indicate the relative angular orientation of the two point quadrupoles, as discussed above. Note that no long range correction is necessary for this interaction type if the fluid is isotropic.

Point dipole – point charge interactions

The interaction between a point dipole with moment μ_i and a point charge q_j at a distance r_{ij} is given by [Hirschfelder et al. 1954; Lorrain et al. 1995]

$$u_{ij}^{Dq}\left(r_{ij}, \theta_i, \mu_i, q_j\right) = -\frac{1}{4\pi\epsilon_0}\frac{\mu_i q_j}{r_{ij}^2}\cos\theta_i\ . \tag{5.5}$$

Here, θ_i is the angle between the distance vector of the point charge and the orientation vector of the point dipole, as illustrated in Figure 5.1.

Point quadrupole – point charge interactions

The interaction potential of a linear point quadrupole Q_i with a point charge q_j at a distance r_{ij} is given by Allen 2002; Gray et al. 1984

$$u_{ij}^{Qq}\left(r_{ij}, \theta_i, Q_i, q_j\right) = \frac{1}{4\pi\epsilon_0}\frac{Q_i q_j}{4r_{ij}^3}\left(3\cos^2\theta_i - 1\right)\ , \tag{5.6}$$

where θ_i is the angle between the distance vector of the point charge and the orientation vector of the point quadrupole, as illustrated in Figure 5.1.

Point quadrupole – point dipole interactions

The interaction between a linear point quadrupole with moment Q_i and a point dipole with moment μ_j is given by Gray et al. 1984; Hirschfelder et al. 1954

$$\begin{aligned} u_{ij}^{Q\mu}\left(r_{ij},\theta_i,\theta_j,\phi_{ij},Q_i,\mu_j\right) &= \frac{1}{4\pi\epsilon_0}\frac{3}{2}\frac{Q_i\mu_j}{r_{ij}^4}\left(\cos\theta_i-\cos\theta_j\right) \\ & \left(1+3\cos\theta_i\cos\theta_j-2\cos\phi_{ij}\sin\theta_i\sin\theta_j\right)\ , \end{aligned} \tag{5.7}$$

where the angles θ_i, θ_j and ϕ_{ij} indicate the relative angular orientation of the point dipole j and the point quadrupole i, as shown in Figure 5.1.

5.4.3 Combination rules

For pure components, the interactions between two different LJ sites is described by the Lorentz-Berthelot combination rules [Berthelot 1898; Lorentz 1881]. For mixtures, the combination rules are extended to the modified Lorentz-Berthelot rules, which include two additional parameters η and ξ to describe the interactions between LJ sites of unlike molecules [Fischer et al. 1989]

$$\sigma_{ij} = \eta\frac{\sigma_i+\sigma_j}{2}\ , \tag{5.8}$$

$$\epsilon_{ij} = \xi\sqrt{\epsilon_i\epsilon_j}\ . \tag{5.9}$$

The two parameters scale all LJ interaction sites between molecules of different components equally. Arbitrary modifications of the combination rules are possible. $ms2$ allows the specification of η and ξ for every molecule pair independently and thus the free parameterization of the combination rules.

5.5 Cutoff & long range correction

The reaction field method and Ewald summation are implemented in $ms2$ as long range correction schemes as briefly described in the following.

5.5.1 Reaction field

The truncation of interactions between first and second order electrostatic sites leads to errors that need to be corrected. The reaction field method [**nymand**; Frenkel 2002] is implemented in $ms2$ for this task, being widely used [Steinhauser 1982; Spoel et al. 1998]. Its advantages are accuracy and stability, while requiring little computational effort compared to other techniques like Ewald summation [Ewald 1921]. However, it is limited to electro-neutral systems, i. e. uncharged molecules. The basic assumption of the reaction field method is that the system is sufficiently large so that tinfoil boundary conditions ($\epsilon_s\rightarrow\infty$) are applicable without a loss of accuracy. This is the case for $N\geq 500$ [Lisal et al. 1997; Garzon et al. 1995; Lisal et al. 2001; Jedlovszky et al. 1999].

All dipoles within the cut-off radius r_c polarize the fluid surrounding the cut-off sphere, which is modeled as a dielectric continuum with a relative permittivity ϵ_s. The polarization gives rise to a homogeneous electric field within the cut-off radius, called the reaction field $\boldsymbol{E}_i^\mathrm{RF}$ of magnitude

$$\boldsymbol{E}_i^\mathrm{RF}=\frac{1}{4\pi\epsilon_0}\frac{2\left(\epsilon_s-1\right)}{2\epsilon_s+1}\frac{1}{r_\mathrm{c}^3}\sum_{b=1}^{m}\sum_{\substack{j=1\\ r_{ij}<r_\mathrm{c}}}^{N_b}\boldsymbol{\mu}_{b,j}\ . \tag{5.10}$$

Note that all m dipoles with moment $\boldsymbol{\mu}$ on all N_b molecules within the cut-off sphere have to be summed up. The reaction field acting outside of the cut-off sphere interacts with a dipole $\boldsymbol{\mu}_i$ at the center of the cut-off sphere by [Allen 2002; Boettcher et al. 1973]

$$u_i^{\mu\mathrm{RF}} = -\frac{1}{2}\boldsymbol{\mu}_i \boldsymbol{E}_i^{\mathrm{RF}} \; , \tag{5.11}$$

where $u_i^{\mu\mathrm{RF}}$ is its contribution to the potential energy. Eq. (5.11) makes use of the assumption that the system is sufficiently large. The reaction field method can also be applied to correct for sets of point charges, as long as they add up to a total charge of zero. Therefore, a point charge distribution is reduced to a dipole vector $\boldsymbol{\mu}^q$ according to

$$\boldsymbol{\mu}^q = \sum_{i=1}^{n} \boldsymbol{r}_i q_i \; , \tag{5.12}$$

where n denotes the total number of point charges and $\boldsymbol{r}_i$ the position vector of a charge q_i. The resulting dipole $\boldsymbol{\mu}^q$ is transferred into the correction term according to Eqs. (5.10) and (5.11).

5.5.2 Ewald summation

Ewald summation [Allen 2002; Frenkel 2002] was implemented for the calculation of electrostatic interactions between point charges. It extends the applicability of $ms2$ to thermodynamic properties of e. g. ions in solutions. In Ewald summation, the electrostatic interactions according to Coulomb's law are divided into two contributions: short-range and long-range. The short-range term includes all charge-charge interactions at distances smaller than the cut-off radius. The remaining contribution is calculated in Fourier space and only the final value is transformed back into real space. This allows for an efficient calculation of the long-range interactions between the charges. The algorithm is well described in the literature. Currently, some of the newest features, the calculation of Massieu potential derivatives and Hybrid MPI & OpenMP Parallelization for MD, are not available with Ewald summation.

5.6 Computing physical properties

$ms2$ enables for the calculation of a large variety of physical properties without any further packages or post-processing. $ms2$ provides the calculation of many static thermodynamic properties, such as the Helmholtz energy A, chemical potential, heat capacities, speed of sound, pressure, temperature, internal energy etc., transport properties, such as diffusion coefficients, thermal conductivity or sheer viscosity and configurational properties, such as radial distribution function and H-bonding statistics. In this section each physical property that is provided by $ms2$ is described.

5.6.1 Ensembles overview

The following simulation types are currently implemented in $ms2$:

- canonical ensemble (NVT) MC (corresponds to MD with thermostat)
- micro-canonical ensemble (NVE) MC (corresponds to MD)
- isobaric-isothermal ensemble (NpT) MC (corresponds to MD with thermostat and barostat)
- grand equilibrium method (pseudo-μVT) MC (MC only)
- isoenthalpic–isobaric ensemble (NpH)

An ensembles is the set of all theoretically possible configurations (states) under some specific macroscopic conditions. For example, the canonical ensemble NVT is the set of all configurations that fulfills the condition of having the same temperature T, volume V and number of particles N, whereas the isobaric-isothermal ensemble NpT represents a system at constant temperature T, pressure p, and number of particles N. Independent on the specific macroscopic conditions, a Monte Carlo simulation always involves the creation of a trial configuration according to some probability α_{ij}, where i refers to the previous state of the system, while j refers to the new one generated with α_{ij} probability from i. Another probability, a_{ij}, determines that the newly generated state j is actually accepted as a proper configuration of the specific ensemble or not. Thus, the transition probability between state i and j is $\pi_{ij} = \alpha_{ij} a_{ij}$. The transition probability for state i depends only on the previous state j. This property is called the Markov chain property; the sequence of configurations created by this way is defined as the Markov chain. It can be shown that if a Markov chain satisfies the condition $P_i\pi_{ij} = P_j\pi_{ji}$ (for every state i and j) of *microscopic reversibility* then the probability of every configuration in the Markov chain will converge to the arbitrarily chosen set of probabilities $\mathbf{P} = [P_1, P_2, ..., P_i, ...]$ as the length of the chain increases. Once $\mathbf{P}$ is specified, the microscopic reversibility condition $P_i\pi_{ij} = P_j\pi_{ji}$ can be solved and has potentially more than one solution for a_{ij}. A solution for a_{ij} is called the acceptance criterion of the ensemble. The generation of the next potential configuration is done such a way that α is symmetric, i.e. $\alpha_{ij} = \alpha_{ji}$. This way α_{ij} and α_{ji} cancel each other in the equation. It is sensible to specify $\mathbf{P} = [P_1, P_2, ..., P_i, ...]$ as the set of actual probability of occurrence of the configurations $1, 2, ..., i, ...$ in the ensemble.
With the exception of the NVE ensemble, not every theoretically possible configuration of an ensemble has the same probability of occurrence. In fact, only a small subset of configurations is truly relevant even though the set of all theoretically possible configurations of an ensemble is extremely large. Although, the analytical form of P_i is known for an ensemble, it's numerical value is not. Therefore, the way of calculating a macroscopic property Z as an *ensemble average* using the standard statistical formula for the expected value $< Z >= \sum Z_i P_i$ (for every possible i), where Z_i is the value of property Z for configuration i, is not feasible. However, it can be shown that if configurations are selected randomly (*Monte Carlo sampling*) and selected according to their actual probability of occurrence P_i at the same time (*importance sampling*) then the expected value can be calculated as $< Z >= \sum Z_i/n$ (assuming the number of sampled configurations n is large enough). A Markov chain satisfying the microscopic reversibility criterion $P_i\pi_{ij} = P_j\pi_{ji}$ fulfills exactly this condition, because it selects n configurations according to their actual probability of occurrence P_i.

For the micro-canonical ensemble, the MC acceptance criterion for molecular translation and rotation is

$$\min\left(1, \frac{(E-U_j)^{(F-2)/2}\theta(\mathrm{E}-\mathrm{U_j})}{(E-U_i)^{(F-2)/2}\theta(\mathrm{E}-\mathrm{U_i})}\right), \tag{5.13}$$

where E is the total *hamiltonian* of the system that is a sum of the total kinetic K and potential energy U of the system, F is the sum of the individual molecular degrees of freedom in the system, and θ is the unit step function. The MC acceptance criterion for the canonical ensemble for molecular translation and rotation is

$$\min\left(1, \exp(-\beta(U_j - U_i))\right), \tag{5.14}$$

where $\beta = 1/(k_\mathrm{B}T)$.

For the NpT ensemble, the MC acceptance criterion for molecular translation and rotation as well as volume change is

$$\min\left(1, \exp(-\beta(U_j - U_i + p(V_j - V_i)) + N\ln(V_j/V_i))\right). \tag{5.15}$$

5.6.2 Statistical uncertainties

$ms2$ calculates the thermodynamic properties from the trajectories (MD) or Markov Chains (MC) on the fly. The results are written to a file with a specified frequency during the course of the simulation. The statistical uncertainties of all results are estimated using the block averaging method according to Flyvbjerg and Petersen [Flyvberg et al. 1989] and the error propagation law.

5.6.3 Time-independent properties from Helmholtz energy derivatives (NVT, NVE)

Standard textbook approaches in the molecular simulation literature indicate specific statistical mechanical ensembles for particular thermodynamic properties. E.g., properties originating from differentiation with respect to temperature at constant pressure, such as the isothermal compressibility, are most likely considered to be candidates for the NpT ensemble, while for a property such as the isochoric heat capacity, the NVT ensemble is considered to be a natural choice. It is true that certain properties have simpler statistical analogues in certain ensembles and may be difficult to derive in other ensembles. Fundamentally, however, the above mentioned philosophy is misleading. In principle, any thermodynamic property is accessible in any ensemble. The formalism proposed by Lustig [Lustig 2011; Lustig 2012] allows for the calculation of an arbitrary number of derivatives

$$A^{\mathrm{r}}_{nm} = (1/T)^n \rho^m \frac{\partial^{n+m} f^{\mathrm{r}}(T,\rho)/(RT)}{\partial(1/T)^n \partial\rho^m} \tag{5.16}$$

for $n > 0$ or $m > 0$ in the NVT and NVE ensembles, with the molar residual Helmholtz energy a^{res}, the temperature T, the density ρ, the molar gas constant R. It can be shown that the total $a/(RT)$ can be additively separated into an ideal $a^{\mathrm{id}}/(RT)$ and a residual part $a^{\mathrm{res}}/(RT)$. The ideal part by definition corresponds to the value of $a/(RT)$ when no intermolecular interactions are at work. $a^{\mathrm{id}}/(RT)$ consists of an exclusively temperature and an exclusively density dependent part. The former has a non-trivial temperature dependence, whereas the latter has a trivial density dependence $\ln(\rho/\rho_{\mathrm{ref}})$.
The approach is based on the fact that the fundamental equation of state contains the complete thermodynamic information about a system, which can be expressed in terms of various thermodynamic potentials, e. g. internal energy $E(N,V,S)$, enthalpy $H(N,p,S)$, Helmholtz energy $A(N,V,T)$ or Gibbs energy $G(N,p,T)$, with number of particles N, volume V, pressure p, temperature T and entropy S. These representations are equivalent in the sense that any other thermodynamic property is essentially a combination of derivatives of the chosen form with respect to its independent variables. The form $a/(RT)(N,V,1/T)$, known as the Massieu potential, is preferable due to practical reasons. Namely, neither its independent variables, nor its derivatives with respect these variables, involve entropic properties. Moreover, some of its derivatives are directly related to measurable properties.
Since a is a thermodynamic potential, any other static thermodynamic property is a combination of its partial derivatives. The molar internal energy u, pressure p, molar enthalpy h, molar Gibbs energy g, and molar isochoric heat capacity c_{v} are simple functions

$$\frac{u}{RT} = A^{\mathrm{o}}_{10}(T) + A^{\mathrm{r}}_{10}, \tag{5.17}$$

$$\frac{p}{\rho RT} = 1 + A^{\mathrm{r}}_{01}, \tag{5.18}$$

$$\frac{1}{RT}\left(\frac{\partial p}{\partial \rho}\right)_T = 1 + 2A^{\mathrm{r}}_{01} + A^{\mathrm{r}}_{02}, \tag{5.19}$$

$$\frac{1}{\rho R}\left(\frac{\partial p}{\partial T}\right)_{\rho} = 1 + A^{\mathrm{r}}_{01} - A^{\mathrm{r}}_{11}, \tag{5.20}$$

$$\frac{h}{RT} = 1 + A^{\mathrm{r}}_{01} + A^{\mathrm{o}}_{10}(T) + A^{\mathrm{r}}_{10}, \tag{5.21}$$

$$\frac{g}{RT} = 1 + A^{\mathrm{r}}_{01} + A^{\mathrm{o}}_{00} + A^{\mathrm{r}}_{00}, \tag{5.22}$$

$$\frac{c_v}{R} = -A^{\mathrm{o}}_{20}(T) - A^{\mathrm{r}}_{20}, \tag{5.23}$$

while molar isobaric heat capacity c_{p}, and speed of sound w are non-linear combinations of derivatives

$$\frac{c_p}{R} = -A^{\mathrm{o}}_{20}(T) - A^{\mathrm{r}}_{20} + \frac{(1 + A^{\mathrm{r}}_{01} - A^{\mathrm{r}}_{11})^2}{1 + 2A^{\mathrm{r}}_{01} + A^{\mathrm{r}}_{02}}, \tag{5.24}$$

$$\frac{Mw^2}{RT} = 1 + 2A^{\mathrm{r}}_{01} + A^{\mathrm{r}}_{02} - \frac{(1 + A^{\mathrm{r}}_{01} - A^{\mathrm{r}}_{11})^2}{A^{\mathrm{o}}_{20}(T) + A^{\mathrm{r}}_{20}}, \tag{5.25}$$

where M is the molar mass, $A^{\mathrm{r}}_{nm} = A^{\mathrm{r}}_{nm}(T,\rho)$, and $A^{\mathrm{o}}_{nm} = A^{\mathrm{o}}_{nm}(T,\rho) = (1/T)^n\rho^m \cdot \partial^{n+m}[\alpha^{\mathrm{o}}(T) + \alpha^{\mathrm{o}}(\rho)]/\partial(1/T)^n/\partial\rho^m = A^{\mathrm{o}}_{nm}(T) + A^{\mathrm{o}}_{nm}(\rho)$. Note that the ideal part

- $A^{\mathrm{o}}_{nm}(T,\rho) = 0$, for $n > 0$ and $m > 0$,
- $A^{\mathrm{o}}_{nm}(T,\rho) = A^{\mathrm{o}}_{nm}(T) + 0$, for $n > 0$ and $m = 0$,
- $A^{\mathrm{o}}_{nm}(T,\rho) = 0 + (-1)^{1+m}$, for $n = 0$ and $m > 0$.

A more complete list of thermodynamic properties can be found in ref. [Span 2000]. For the NVE ensemble, there are two sets of A^{r}_{nm} output values in $ms2$. Each set corresponds to a different statistical mechanical entropy definition.The numerical values for these definitions should converge to the same number as the system size increases and they should be identical in the thermodynamic limit ($N \to \infty$).

The A^{r}_{nm} derivatives are determined from fluctuation formulas. These fluctuation formulas require the ensemble average of U^n and $\partial^m U/\partial V^m$ with respect to the applied molecular interaction pair potential. Thus the explicit analytical expressions for U^n and $\partial^m U/\partial V^m$ have to be determined analytically beforehand [Lustig 2011; Lustig 2012]. The general formula for $\partial^m U/\partial V^m$ can be found in Ref. [Lustig 1994]. For common molecular interaction pair potentials, like the Lennard-Jones potential [Allen 2002], describing repulsive and dispersive interactions, or Coulomb's law, describing electrostatic interactions between point charges, the explicit analytical formulas can be obtained straightforwardly. The mathematical form of the long range correction (LRC) for U and $\partial^m U/\partial V^m$ depends on the molecular interaction potential and the cut-off method (site-site or center-of-mass cut-off mode) applied. For the Lennard-Jones potential, the LRC scheme was well described in the literature for both the site-site [Lustig 2011; Meier et al. 2006] and the center of mass cut-off mode [Lustig 1994; Lustig 1998]. For the usual implementation of the reaction field method [Barker et al. 1973; Omelyan 1996], the LRC for $\partial^m U/\partial V^m$ can be obtained straightforwardly. However, practical applications show that the electrostatic LRC of $\partial U/\partial V$ and $\partial^2 U/\partial V^2$ can be neglected in case of systems for which the reaction field method is an appropriate choice. E.g., the contribution of the electrostatic LRC for a liquid system ($T = 298$ K and $\rho = 45.86$ mol/l) consisting of only 200 water and 50 methanol molecules with a very short cut-off radius of 20% of the edge length of the simulation volume is still $<< 1\%$ for both $\partial U/\partial V$

and $\partial^2 U/\partial V^2$.

5.6.4 Static properties from conventional fluctuation formulas (NpT)

Several static thermodynamic properties have simple statistical analogues in certain ensembles. In the NpT ensemble, the second derivatives of the Gibbs energy, namely the residual isobaric heat capacity c_p^{res}, the isothermal compressibility β_T and the volume expansivity α_p are functions of ensemble fluctuations [Allen 2002]. The residual isobaric heat capacity c_p^{res} is related to fluctuations of the residual enthalpy H^{res}

$$c_p^{\mathrm{res}} = \frac{1}{N}\left(\frac{\partial H^{\mathrm{res}}}{\partial T}\right)_p = \frac{1}{k_{\mathrm{B}}\left(NT\right)^2}\left(\left\langle (H^{\mathrm{res}})^2\right\rangle - \left\langle H^{\mathrm{res}}\right\rangle^2\right) . \tag{5.26}$$

To obtain the total isobaric heat capacity c_p, the solely temperature dependent ideal gas contribution $c_p^{\mathrm{id}}\left(T\right)$ has to be added. This ideal property is accessible e. g. by quantum chemical methods and can often be found in data bases [Rowley 2003].

An analogous relationship links the isothermal compressibility β_T to volume fluctuations in the NpT ensemble

$$\beta_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T = \frac{1}{k_{\mathrm{B}}T\left\langle V\right\rangle}\left(\left\langle V^2\right\rangle - \left\langle V\right\rangle^2\right) . \tag{5.27}$$

The partial derivative of the residual enthalpy with respect to the pressure at constant temperature $(\partial H^{\mathrm{res}}/\partial p)_T$ is linked to fluctuations of volume and residual internal energy

$$\left(\frac{\partial H^{\mathrm{res}}}{\partial p}\right)_T = V - \frac{N}{T}\left(\left\langle U^{\mathrm{res}}V\right\rangle - \left\langle U^{\mathrm{res}}\right\rangle\left\langle V\right\rangle + p\left(\left\langle V^2\right\rangle - \left\langle V\right\rangle^2\right)\right) , \tag{5.28}$$

and the volume expansivity α_p again to volume and residual enthalpy fluctuations

$$\alpha_p = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p = \frac{N}{T^2\left\langle V\right\rangle}\left(\left\langle H^{\mathrm{res}}V\right\rangle - \left\langle H^{\mathrm{res}}\right\rangle\left\langle V\right\rangle\right) . \tag{5.29}$$

The speed of sound w is the velocity of a sound wave traveling through an elastic medium. It can be given by the isothermal compressibility β_T, the volume expansivity α_p, the isobaric heat capacity c_p and the temperature T by

$$w = \left(\frac{1}{M\left(\beta_T\rho - T\alpha_p^2/c_p\right)}\right)^{0.5} , \tag{5.30}$$

where M is the molar mass. In $ms2$, the speed of sound is calculated both for pure components and mixtures in the NpT ensemble.

5.6.5 Transport properties

In $ms2$, transport properties are determined via equilibrium MD simulations by means of the Green-Kubo formalism. This formalism offers a direct relationship between transport coefficients and the time integral of the autocorrelation function of the corresponding fluxes.

An extended time increment is defined in $ms2$ for the calculation of the fluxes, the autocorrelation functions and their integrals, which may be significantly times longer than the MD time step. As a consequence, the memory demand for the autocorrelation functions is reduced and the restart file, which contains the current state of the autocorrelation functions and time integrals, is accordingly smaller.

Diffusion coefficients

The self-diffusion coefficient D_i is related to the mass flux of single molecules of component i within a fluid. Therefore, the relevant Green-Kubo expression is based on the individual molecule velocity autocorrelation function [Gubbins 1972]

$$D_i = \frac{1}{3N_i} \int_0^\infty \mathrm{d}t \left\langle \mathbf{v}_k\left(t\right) \cdot \mathbf{v}_k\left(0\right) \right\rangle \ , \tag{5.31}$$

where $\mathbf{v}_k\left(t\right)$ is the center of mass velocity vector of molecule k at some time t. Eq. (5.31) is an average over all N_i molecules of component i in the ensemble, since all contribute to the self-diffusion coefficient. The transport diffusion in a mixture is related to the phenomenological coefficients L_{ij} which can be calculated from the Green-Kubo expression based on the net velocity autocorrelation function

$$L_{ij} = \frac{1}{3N} \int_0^\infty \mathrm{d}t \left\langle \sum_{k=1}^{N_i} \mathbf{v}_{i,k}\left(0\right) \cdot \sum_{l=1}^{N_j} \mathbf{v}_{j,l}\left(t\right) \right\rangle \ . \tag{5.32}$$

where N is the total number of molecules. Because of Onsager's reciprocity theorem, $L_{ij} = L_{ji}$ for $i \neq j$. In this context, the Maxwell-Stefan diffusion coefficient for binary mixtures is given by [Krishna et al. 2005]

$$Ð_{ij} = \frac{x_j}{x_i}\Lambda_{ii} + \frac{x_i}{x_j}\Lambda_{jj} - \Lambda_{ij} - \Lambda_{ji} \ . \tag{5.33}$$

From the expressions above, the collective character of the Maxwell-Stefan diffusion coefficient is evident. This leads to significantly less data for a given system size and time step and therefore to larger statistical uncertainties than in case of the self-diffusion coefficient. Note that equations for the Maxwell-Stefan diffusion coefficient are implemented in $ms2$ both for binary and ternary mixtures.

Shear viscosity

The shear viscosity η, as defined by Newton's "law" of viscosity, is a measure of the resistance of a fluid to a shearing force [Hoheisel 1994]. It is associated with the momentum transport under the influence of velocity gradients. Hence, the shear viscosity can be related to the time autocorrelation function of the off-diagonal elements of the stress tensor $\mathbf{J}_p$ [Gubbins 1972]

$$\eta = \frac{1}{Vk_\mathrm{B}T} \int_0^\infty \mathrm{d}t \left\langle J_p^{xy}\left(t\right) \cdot J_p^{xy}\left(0\right) \right\rangle \ . \tag{5.34}$$

Five independent terms of the stress tensor J_p^{xy}, J_p^{xz}, J_p^{yz}, $(J_p^{xx} - J_p^{yy})/2$ and $(J_p^{yy} - J_p^{zz})/2$ were considered to improve statistics Alfe et al. 1988. The component J_p^{xy} of the microscopic stress tensor $\mathbf{J}_p$ is given in terms of the molecular positions and velocities by [Hoheisel 1994]

$$J_p^{xy} = \sum_{i=1}^{N-1} m v_i^x v_i^y - \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \sum_{k=1}^{n} \sum_{l=1}^{n} r_{ij}^x \frac{\partial u_{ij}}{\partial r_{kl}^y} \ . \tag{5.35}$$

Here, the lower indices l and k indicate the n interaction sites of a molecule and the upper indices x and y denote the spatial vector components, e.g. for velocity v_i^x or site-site distance r_{ij}^x. The first term of Eq. (5.35) is the kinetic energy contribution and the second term is the potential energy contribution to the shear viscosity. Consequently, the Green-Kubo integral (5.34) can be decomposed into three parts, i. e. the solely kinetic energy contribution, the solely potential energy contribution and the mixed kinetic-potential energy contribution [Hoheisel 1994].

Bulk viscosity
The bulk viscosity η_v refers to the resistance to dilatation of an infinitesimal volume element at constant shape [**Barton1998**]. The bulk viscosity can be calculated by integration of the time-autocorrelation function of the diagonal elements of the stress tensor and an additional term that involves the product of pressure p and volume V, which does not appear in the shear viscosity, cf. Eq. (5.34). In the NVE ensemble, the bulk viscosity is given by [Gubbins 1972; Steele 1969]

$$\eta_v = \frac{1}{Vk_\mathrm{B}T}\int_0^\infty \mathrm{d}t \left\langle \left(J_p^{xx}(t) - pV(t)\right)\cdot\left(J_p^{xx}(0) - pV(0)\right)\right\rangle \ . \tag{5.36}$$

The diagonal component J_p^{xx} of the microscopic stress tensor $\mathbf{J}_p$ is defined analogous to Eq. (5.35). The statistics of the ensemble average in Eq. (5.36) can be improved using all three independent diagonal elements of the stress tensor J_p^{xx}, J_p^{yy} and J_p^{zz}. Eqs. (5.35) and (5.36) can directly be applied for mixtures.

Thermal conductivity
The thermal conductivity λ is given by the autocorrelation function of the elements of the microscopic heat flow J_q^x

$$\lambda = \frac{1}{Vk_BT^2}\int_0^\infty dt\ \left\langle J_q^x(t)\cdot J_q^x(0)\right\rangle. \tag{5.37}$$

In mixtures, energy transport and diffusion occur in a coupled manner, thus, the heat flow for a mixture of n components is given by [Evans et al. 1978]

$$\begin{aligned}\mathbf{J}_q \quad &= \quad \frac{1}{2}\sum_{i=1}^{n}\sum_{k=1}^{N_i}\left[m_i^k\left(v_i^k\right)^2 + \mathbf{w}_i^k\mathbf{I}_i^k\mathbf{w}_i^k + \sum_{j=1}^{n}\sum_{l\neq k}^{N_j} u\left(r_{ij}^{kl}\right)\right]\cdot\mathbf{v}_i^k \\ &- \quad \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{N_i}\sum_{l\neq k}^{N_j}\boldsymbol{r}_{ij}^{kl}\cdot(\mathbf{v}_i^k\cdot\frac{\partial u\left(r_{ij}^{kl}\right)}{\partial \boldsymbol{r}_{ij}^{kl}} + \mathbf{w}_i^k\boldsymbol{\Gamma}_{ij}^{kl}) - \sum_{i=1}^{n} h_i\sum_{k=1}^{N_i}\mathbf{v}_i^k\ ,\end{aligned} \tag{5.38}$$

where $\mathbf{w}_i^k$ is the angular velocity vector of molecule k of component i and $\mathbf{I}_i^k$ its matrix of angular momentum of inertia. $u\left(r_{ij}^{kl}\right)$ is the intermolecular potential energy and $\boldsymbol{\Gamma}_{ij}^{kl}$ is the torque due to the interaction of molecules k and l. The indices i and j denote the components of the mixture. h_i is the partial molar enthalpy. For the calculation of the thermal conductivity of mixtures h_i has to be specified as an input in the $ms2$ parameter file for each component and can be calculated in advance from NpT simulations.

Electric conductivity
The electric conductivity σ is a measure for the flow of ions in solution. The Green-Kubo formalism [Gubbins 1972] offers a direct relationship between σ and the time-autocorrelation function of the electric current flux $\boldsymbol{j_e}(t)$ [Hansen et al. 1986]

$$\sigma = \frac{1}{3Vk_\mathrm{B}T}\int_0^\infty \ \langle \boldsymbol{j_e}(t)\cdot\boldsymbol{j_e}(0)\rangle \mathrm{d}t\ , \tag{5.39}$$

where k_B is Boltzmann's constant. The electric current flux is defined by the charge q_k of ion k and its velocity vector $\boldsymbol{v}_k$ according to

$$\boldsymbol{j_e}(t) = \sum_{k=1}^{N_j} q_k\cdot\boldsymbol{v}_k(t)\ , \tag{5.40}$$

where N_j is the number of molecules of component j in solution. Note that only the ions in the solution have to be considered, not the electro-neutral molecules. For better statistics, σ is sampled over all independent spatial elements of $\boldsymbol{j_e}(t)$.

5.6.6 Configurational properties

Radial distribution function
The radial distribution function (RDF) $g(r)$ is a measure for the microscopic structure of matter. It is defined by the local number density around a given position within a molecule $\rho^L(r)$ in relation to the overall number density $\rho = N/V$

$$g(r) = \frac{\rho^L(r)}{\rho} = \frac{1}{\rho}\frac{\mathrm{d}N(r)}{\mathrm{d}V} = \frac{1}{4\pi r^2 \rho}\frac{\mathrm{d}N(r)}{\mathrm{d}r}. \tag{5.41}$$

Therein, $\mathrm{d}N(r)$ is the differential number of molecules in a spherical shell volume element $\mathrm{d}V$, which has the width $\mathrm{d}r$ and is located at the distance r from the regarded position. $g(r)$ can be evaluated for every molecule of a given species.

In the present release of *ms*2, the RDF can be calculated during MD and MC simulation runs for pure components and mixtures on the fly. The RDF is sampled between all Lennard-Jones sites. In order to evaluate RDF for arbitrary positions, say point charge sites, superimposed dummy Lennard-Jones sites with the parameters $\sigma = \epsilon = 0$ have to be introduced in the potential model file by the user.

The residence time τ_j defines the average time span that a molecule of component j remains within a given distance r_{ij} around a specific molecule i. It is given by the autocorrelation function

$$\tau_j = \int_{t=0}^{\infty} \left\langle \frac{1}{n_{ij}(0)} \sum_{k=1}^{n_{ij}(0)} \Theta_k(t)\Theta_k(0) \right\rangle \mathrm{d}t \ , \tag{5.42}$$

where t is the time, $n_{ij}(0)$ the solvation number around molecule i at $t = 0$ and Θ is the Heaviside function, which yields unity, if the two molecules are within the given distance, and zero otherwise. Following the proposal of Impey et al. Impey et al. 1983, the residence time explicitly allows for short time periods during which the distance between the two molecules exceeds r_{ij}. Also, the solvation number n_{ij} is evaluated on the fly

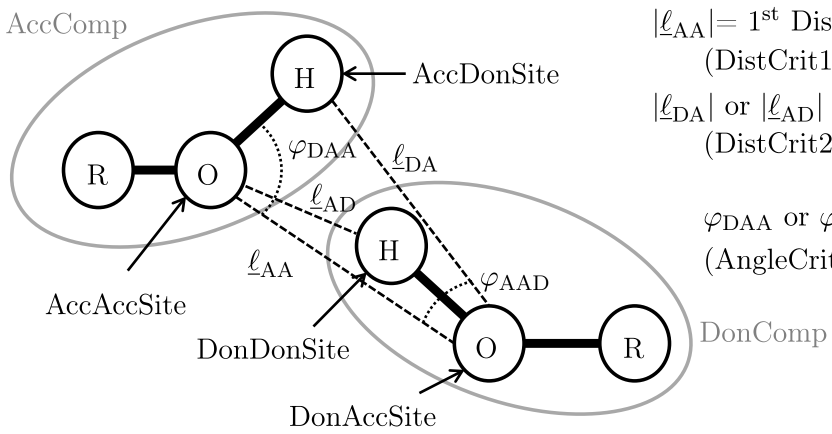
$$n_{ij} = 4\pi\rho_j \int_0^{r_{\min}} r^2 g_{ij}(r)\mathrm{d}r \ , \tag{5.43}$$

where ρ_j is the number density of component j and $r_{\min}$ is the distance up to which the solvation number is calculated.

Hydrogen bonding

A geometric criterion is implemented in *ms*2 to identify H-bonding networks in fluids. Although there is no unambiguous definition of what an H-bond between two interacting molecules is, it is commonly characterized by the following two criteria [Langenbach et al. 2015; Arunan et al. 2011]:

- The interaction between two sites of an H-bond is highly directional and short ranged.
- One donor always interacts with one acceptor.



$|\underline{\ell}_{\mathrm{AA}}|$= 1st Distance criterion in Å (DistCrit1)

$|\underline{\ell}_{\mathrm{DA}}|$ or $|\underline{\ell}_{\mathrm{AD}}|$ = 2nd Distance criterion in Å (DistCrit2)

φ_{DAA} or φ_{AAD} = Angle criterion in ° (AngleCrit)

$$|\underline{\ell}_{\mathrm{AA}}| \geq |\underline{r}_{\mathrm{AccAccSite}} - \underline{r}_{\mathrm{DonAccSite}}| \tag{5.44}$$

$$|\underline{\ell}_{\mathrm{DA}}| \geq |\underline{r}_{\mathrm{AccDonSite}} - \underline{r}_{\mathrm{DonAccSite}}| \tag{5.45}$$

$$|\underline{\ell}_{\mathrm{AD}}| \geq |\underline{r}_{\mathrm{AccAccSite}} - \underline{r}_{\mathrm{DonDonSite}}| \tag{5.46}$$

$$\varphi_{\mathrm{DAA}} \leq \arccos \left| \frac{(\underline{r}_{\mathrm{AccDonSite}} - \underline{r}_{\mathrm{AccAccSite}}) \cdot \underline{\ell}_{\mathrm{AA}}}{|\underline{r}_{\mathrm{AccDonSite}} - \underline{r}_{\mathrm{AccAccSite}}| \cdot |\underline{\ell}_{\mathrm{AA}}|} \right| \tag{5.47}$$

$$\varphi_{\mathrm{AAD}} \leq \arccos \left| \frac{(\underline{r}_{\mathrm{DonAccSite}} - \underline{r}_{\mathrm{DonDonSite}}) \cdot \underline{\ell}_{\mathrm{AA}}}{|\underline{r}_{\mathrm{DonAccSite}} - \underline{r}_{\mathrm{DonDonSite}}| \cdot |\underline{\ell}_{\mathrm{AA}}|} \right| \tag{5.48}$$

Figure 5.2: Schematic of geometric H-bond criteria.

The implemented geometric criterion of [Haughney et al. 1987] identifies, whether an interaction between two sites is accounted for a H-bond.

> A molecule is the H-donor to another molecule, if the distance between the donor and the acceptor is smaller than ℓ_{DA} or ℓ_{AD}, the distance between two acceptors is smaller than ℓ_{AA} and the angle between the acceptor-donor axis and the acceptor-acceptor axis is smaller than φ_{DAA} or φ_{AAD}. The original numeric values suggested by *Haughney et al.* are ℓ_{AA} =3.5 Å, ℓ_{DA} =2.6 Å or ℓ_{DA} =2.6 Å and φ_{DAA} =30° or φ_{AAD} =30°. They have to be specified by the user as described in the following.

As the H-bonding analysis is implemented in *ms*2, charge-charge-charge configurations relating to two components (which can be one and the same) can be sampled. Oxygen and Hydrogen atoms do not necessarily have to be part of these structures. The implemented geometric criterion evaluates the triangle between three charge sites of two molecules – either two sites in the acceptor molecule or two sites in the donor molecule – and compares these with the criterion thresholds.

The user has to specify the geometric relations in the `*.par` file according to the simulated components as follows: First, the keyword `NHBondCriteria = '#h-bonds'` has to be set to the integer value `#h-bonds` according to the number of different H-bonds that shall be investigated during the simulation. This keyword has to appear under the components section in the `*.par` file. `#h-bonds` also sets the number of lines that *ms*2 expects in the `*.par` file, each to have a complete set of nine parameters that declare a certain H-bond. These nine parameters are:
AccComp, AccAccSite, AccDonSite, DonComp, DonAccSite, DonDonSite, DistCrit1, DistCrit2 and AngleCrit. Fig. 5.2 visualises these parameters. The first six parameters declare which charge – charge

interactions of which component are to be considered by *ms*2. The acceptor and donor sites listed in the `*.par` file have to be point charges listed in the `*.pm` file. AccComp and DonComp are the numbers of the components that are supposed to be the acceptor and donor of the H-bond, respectively. Both have to be integer values according to the order of the components listed in the `*.par` file. The four charge sites that participate in the H-bond have to be set by the user as AccAccSite, AccDonSite, DonAccSite and DonDonSite. They have to be integer values according to the order of the charge sites listed in the `.*pm` file. AccAccSite and AccDonSite are the acceptor and donor charge sites in the component that is acting as the acceptor molecule in the H-bond while DonAccSite and DonDonSite are the acceptor and donor charge-sites in the donor molecule. As mentioned above, only three charge sites are needed for the criterion, thus either the AccDonSite or the DonDonSite has to be set to `0` by the user as can be seen in table 5.2. This is equivalent with defining either ℓ_{DA} and φ_{DAA} or ℓ_{AD} and φ_{AAD}.

The last three numeric parameters in table 5.2 are the actual geometric lengths ℓ_{AA}, ℓ_{AD} or ℓ_{DA} and angles φ_{DAA} or φ_{AAD} as described in [Haughney et al. 1987]. *ms*2 always expects these values to be specified in Ångstrom and degree, respectively, independent of the settings of the `Units` keyword in the `*.par` file.

Table 5.2 gives two examples for a complete parameter set to be specified in a `*.par` file. During simulation, *ms*2 lists the analysis of the H-bonding in the `*.rav` file. Following the criterion by *Haughney et al.*, *ms*2 calculates and itemizes the number N_i of molecules with $i = 0, 1, 2, 3$ or more hydrogen bonds. In the `*.rav` file these numbers are listed according to the component combination. E.g. the `N3_( 1, 2, 2, 3)` column lists the number of molecules which correspond to component 1 and are bonded twice with component 2 and once with component 3. As usual for the `*.rav` file, these values are running averages.

Table 5.2: Example for a parameter set for the H-bonding criterion.

AccComp	AccAccSite	AccDonSite	DonComp	DonAccSite	DonDonSite	DistCrit1	DistCrit2	AngleCrit
1	2	0	1	2	3	3.5	2.6	30
1	2	3	1	2	0	3.5	2.6	30

5.6.7 Henry' s law constant

The solubility of solutes in solvents is characterized by the Henry's law constant. In solvents of a given composition, it is solely a function of temperature. The Henry's law constant H_i is related to the residual chemical potential of the solute i at infinite dilution $\mu_i^{\mathrm{res},\infty}$ as defined in e. g. [**shing**] by

$$H_i = \rho k_{\mathrm{B}} T \exp\left(\frac{\mu_i^{\mathrm{res},\infty}}{k_{\mathrm{B}} T}\right) , \tag{5.49}$$

where ρ denotes the solvent density. The residual chemical potential of a component at infinite dilution can often be calculated using Widom's test molecule method. In this case, a simulation of the solvent is performed in the NpT ensemble and the saturated vapor pressure of the solvent, while the solute is introduced as an additional component of the mixture with a molar fraction of zero. Then, the solute is only added in the form of test molecules to determine its chemical potential at infinite dilution. For dense phases, the chemical potential and thus the Henry's law constant can be calculated with the gradual insertion method, if Widom's method fails. The number of solute molecules in the simulation is reduced to one, representing the infinitely diluted molecule.

5.6.8 Second virial coefficient

The second virial coefficient is related to the intermolecular potential by [Gray et al. 1984]

$$B = -2\pi \int_0^\infty \mathrm{d}r_{ij} \left\langle \exp\left(-\frac{u_{ij}\left(r_{ij}, \boldsymbol{\omega}_i, \boldsymbol{\omega}_j\right)}{k_\mathrm{B}T}\right) - 1 \right\rangle_{\boldsymbol{\omega}_i, \boldsymbol{\omega}_j} r_{ij}^2 \,, \tag{5.50}$$

where the $\langle \ldots \rangle$ brackets indicate the average over the orientations $\boldsymbol{\omega}_i$ and $\boldsymbol{\omega}_j$ of two molecules i and j separated by a center of mass distance r_{ij}. The integrand, called Mayer's f-function, is evaluated at numerous distances in an appropriate range. $ms2$ evaluates Mayer's f-function by averaging over numerous random orientations at each radius.

5.6.9 Chemical potential and partial molar properties

The chemical potential of component i can be separated into the temperature T dependent part $\mu_i^\mathrm{id}(T)$ of the ideal gas contribution to the chemical potential of the pure component and the remaining contribution $\mu_\mathrm{i}(T, \mathbf{x}, v) - \mu_i^\mathrm{id}(T)$which is often referred to as the configurational chemical potential. The solely temperature dependent ideal gas contribution $\mu_i^\mathrm{id}(T)$ cancels out in the calculation of phase equilibria and is therefore not implemented in $ms2$. The chemical potential depends on the real substance behavior due to the molecular interactions and can be determined with $ms2$ using two different techniques: Widom's particle insertion method and thermodynamic integration.

Widom's test particle insertion method
Widom's test particle insertion method is a conceptually straightforward approach to calculate the chemical potential of a component i and was presented by Widom Widom 1963. It allows the sampling of the chemical potential with low computational cost, both for pure substances and mixtures. For a three-site Lennard-Jones fluid, the computational demand is shown in Table 5.3. The execution time for the determination of the chemical potential increases roughly linearly with the specified number of test molecules that are sampled.

Table 5.3: Computational demand for the calculation of the chemical potential for a three-center LJ fluid using Widom's test molecule insertion in MD or MC simulation. The calculations were performed on a single core. The simulation time per time step (MD) was 0.043 s and 0.300 s per loop (MC), respectively. The simulations were performed with 500 molecules at a density of 0.23 mol/l. A cut-off radius of 60 Å was employed.

# test molecules	Additional CPU time per MD time step or MC loop / s
1	0.0015
500	0.1925
1000	0.3865
1500	0.5810
2000	0.7750

For the calculation of the chemical potential of component i according to Widom Widom 1963, an additional so-called "test" molecule $N+1$ of component i is inserted into the simulation volume at a random position with a random orientation. At constant temperature and constant pressure or volume its potential energy U_{N+1}, the interaction energy with all other "real" N molecules yields the configurational chemical potential according to

$$\mu_i - \mu_i^\mathrm{id}(T) = -k_\mathrm{B}T \ln \frac{\langle V \exp\left(-U_{N+1}/k_\mathrm{B}T\right)\rangle}{\langle N_i \rangle}, \tag{5.51}$$

where N is the number of particles of the system, N_i is the number of particles for component i, V is the volume and k_B is Boltzmann's constant. The test molecule is removed immediately after the calculation of its potential energy, thus it does not influence the time evolution of the system or the Markov chain, respectively. In contrast to the usual convention, the brackets $<>$ here have a dual meaning. Namely, they stand for either NVT or NpT ensemble average as well as an integral over all the possible position and orientation of the test molecule in the system.

The value of U_{N+1} is highly dependent on the random position or orientation of the test molecule. In addition, the density of the system has a dominant influence on the accuracy of the calculation. For very dense fluids, test molecules almost always overlap with some real molecules, which leads to a potential energy $U_{N+1} \to \infty$ and thus to a vanishing contribution to Eq. (5.51), resulting in poor statistics for the chemical potential or even complete failure of the sampling. Within limits, lower statistical uncertainties of the chemical potential can be achieved by inserting a large number of test molecules into the simulation volume, which leads to an increasing computational demand. Nonetheless, Widom's test particle insertion method certainly fail above a certain density.

Thermodynamic integration

Thermodynamic integration is one solution to overcome the limitations of Widom's test particle insertion method. The idea behind calculating the chemical potential using thermodynamic integration is to avoid insertion of the test particle in system A and perform it in system B for which it is not problematic or practically not required at all. Since the chemical potential is a state variable, its change $\mu_A - \mu_B$ can be calculated using any arbitrary path between state A and B. This path is represented by the parameter λ. It can be shown Frenkel 2002 that the relation between λ and $\mu_A - \mu_B$ is

$$\mu_{A,j} - \mu_{B,j} = \int_{B(\lambda_{\min})}^{A(\lambda_{\max})} \left\langle \frac{\partial U_j(\lambda)}{\partial \lambda} \right\rangle d\lambda \tag{5.52}$$

for particle j of component i. The bracket in this equation denotes NVT or NpT ensemble average and U_j is the potential energy of particle j that must be a part of the system the same way any other particle is. The only difference between j and regular particles is that the interaction energy $U_j(\lambda)$ is scaled between states $A(\lambda_{\max})$ and $B(\lambda_{\min})$ with λ. As long as $\partial U_j(\lambda)/\partial\lambda$ can be calculated analytically and sampled during simulation, the actual way of scaling $U_j(\lambda)$ with λ is completely arbitrary since μ is a state variable. The integration with respect to λ is evaluated by numerically. Assuming that $\mu_{B,j}$ is practically zero or at least can be successfully calculated by Widom's particle insertion method at the end point B using Eq. (5.51), $\mu_{B,j}$ yields the configurational part of the total chemical potential (everything except the temperature dependent part $\mu_i^{id}(T)$ of the ideal gas contribution $\mu_i^{id}(T) + k_{\mathrm{B}}Tln(N_i/<V>)$ of component i). The non-linear scaling $U_j(\lambda) = \lambda^x U_j$ ($\lambda_{\min} \leq \lambda \leq 1$) with an adaptive sampling technique [Fasnacht et al. 2004] was implemented for MC with x and $\lambda_{\min}$ being input parameters. This adaptive technique allows for the sampling of the entire range of $\lambda_{\min} \leq \lambda \leq 1$ in a single MC simulation run with arbitrary resolution for the numerical integration. Besides the standard NVT and NpT ensemble MC moves, the simulation includes changes of λ controlled by a proper MC acceptance criterion ensuring visits at each discrete λ value in the range $\lambda_{\min} \leq \lambda \leq 1$. For MD simulations, changes of λ are carried out also on the fly in a single simulation but without any acceptance criterion.

Accordingly, there are four parameters involved with the present approach:

- `LambdaMin` (the lower boundary of the interval $\lambda_{\min} \leq \lambda \leq 1$, default value is 0.2),
- `NBins` (the number of discrete bins in the interval $\lambda_{\min} \leq \lambda \leq 1$, default value is 100),
- `LambdaStepMax` (defines the change in λ during simulation, which is random value between zero and $\pm$ *LambdaStepMax*, default value is 0.1),
- `LambdaExponent` (exponent x for the non-linear scaling $U_j(\lambda) = \lambda^x U_j$ ($\lambda_{\min} \leq \lambda \leq 1$), its default value is 4.0 but it does not have to be an integer).

One should keep in mind when choosing these parameters that thermodynamic integration always involves larger statistical uncertainties than the equivalent Widom scenario for a state point for which the latter works perfectly. In other words: the larger the contribution of the numerical integration part is compared to the Widom contribution at λ_{min} is, the higher the statistical uncertainty of the calculated chemical potential gets. This means that setting λ_{min} very low or the exponent x very high (e. g. $x = 12$) might not be beneficial in many cases. Moreover, high exponent x values for MD simulations are not advised since the scaling with x at high λ values (where the absolute value of $U_j(\lambda)$ is close to that of the non-scaled state) may result in new $U_j(\lambda)$ energies (thus forces) that non-negligibly interfere with the dynamics of the system when λ is changed during simulation. For MD simulations, a value $x < 1.0$ is possibly a better choice for many systems.

Partial molar properties, namely the volume v_i and enthalpy h_i

$$v_i = \frac{< V^2 \exp(-U_{N+1}/(k_\text{B}T)) >}{< V \exp(-U_{N+1}/(k_\text{B}T)) >} - < V >, \tag{5.53}$$

$$h_i = \frac{< (U_{N+1} + U_N + pV)V \exp(-U_{N+1}/(k_\text{B}T)) >}{< V \exp(-U_{N+1}/(k_\text{B}T)) >} - < U_N + pV >, \tag{5.54}$$

can be straightforwardly be obtained by Widom's test particle insertion [Sindzingre et al. 1987] and their calculation is implemented in *ms*2. U_N is the potential energy of the N particle system, while U_{N+1} is that of the test particle. The brackets $<>$ denote NpT ensemble averages. For thermodynamic integration, the calculation of these properties is currently not implemented.

5.7 Special features

5.7.1 Vapor-liquid equilibrium

The Grand Equilibrium method is a technique to determine the vapor-liquid equilibrium (VLE) of pure substances or mixtures. It is a two-step procedure, where the coexisting phases are simulated independently and subsequently. The specified thermodynamic variables for the VLE are the temperature T and the composition $\mathbf{x}$ of the liquid. In the first step, one NpT simulation of the liquid phase is performed at T , $\mathbf{x}$ and some pressure p_0 to determine the chemical potentials μ_i^l and the partial molar volumes v_i^l of all components, which corresponds to the molar volume in case of a pure substance. If the entropic properties are determined by Widom's test molecule method [Widom 1963], both MD or MC can be used to sample the phase space. A more advanced technique for these properties is implemented in combination with MC [Lyubartsev et al. 1992; Nezbeda et al. 1991]. On the basis of the chemical potentials and partial molar volumes at p_0, first order Taylor expansions can be made for the pressure dependence

$$\mu_i^l(T, \mathbf{x}, p) \approx \mu_i^l(T, \mathbf{x}, p_0) + v_i^l(T, \mathbf{x}, p_0) \cdot (p - p_0) \ . \tag{5.55}$$

Note that *ms*2 yields $\mu_i(T, \mathbf{x}, p) - \mu_i^{\text{id}}(T)$, where $\mu_i^{\text{id}}(T)$ is the temperature dependent part of the ideal gas contribution to the chemical potential of the pure component. $\mu_i^{l,\text{id}}(T)$ does not need to be determined for VLE calculations, because it cancels out when Eq. (5.55) is equated to the corresponding expression for the vapor.

In the second step, one pseudo-μVT simulation [Vrabec et al. 2002] is performed for the vapor phase on the basis of Eq. (5.55) that yields the saturated vapor state point of the VLE. This simulation takes place in a pseudo ensemble in the sense that the specified chemical potentials are not constant, but dependent on the actual pressure of the vapor phase. However, experience based on hundreds of systems [**huang**; Vrabec et al. 2009] shows that the vapor phase simulation rapidly converges to the saturated vapor state point during equilibration so that effectively the equilibrium chemical potentials are specified via the attained vapor pressure. The number of molecules varies during the vapor phase simulation of the grand equilibrium method. Starting from a specified number of molecules in an arbitrarily chosen gaseous state, $ms2$ adjusts the extensive volume V after an equilibration period so that the vapor density does not have to be known in advance. The grand equilibrium method has been widely used and compares favorably to other methods used for vapor liquid equilibrium simulations.

5.7.2 Osmotic pressure for the activity of the solvent

The OPAS method (osmotic pressure for the activity of the solvent) is an alternative to existing methods for determining chemical potentials and related properties in the liquid phase. The method is mainly intended for studying electrolyte solutions, but can in principle also be used for studying mixtures of molecular species. It is only briefly introduced here, for more details, the reader is referred to [Kohns et al. 2016].

The method is based on a direct computation of the osmotic equilibrium between a pure solvent phase and a solution phase. To this end, two planar, semipermeable membranes are introduced in the cubic $ms2$ simulation box. Thereby, the box is divided into an inner compartment, which contains the solution phase, and an outer compartment, which contains the pure solvent phase. The solvent molecules can pass the membrane, but the solute molecules are kept in the inner compartment by applying a force $F = c\Delta z$, if their center of mass position penetrates one of the membranes by a length of Δz. As a result, a pressure difference, called the osmotic pressure $\Pi = p_{\text{in}} - p_{\text{out}}$ is present between the two phases. Assuming an incompressible solvent, it can be shown that this osmotic pressure is directly related to the solvent activity via

$$\ln(a_{\text{solv}}) = -\frac{\Pi}{\rho_{\text{solv}} RT}. \tag{5.56}$$

The density of the pure liquid solvent ρ_{solv} at the temperature T is easily available from separate molecular simulations. The osmotic pressure Π can be sampled as the total force on the membranes per membrane area:

$$\Pi = \frac{\sum_i |F_i|}{A_{\text{tot}}}, \tag{5.57}$$

where the summation runs over all solute molecules, allowing for the calculation of the solvent activity via Eq. (5.56).

OPAS simulations should be carried out in two steps: First, a pseudo-NpT simulation should be performed. When using the OPAS version of the $ms2$ program, the barostat is modified such that the pressure is kept constant at the specified value in the outer compartment, i.e. the pure solvent phase (for details see [Kohns et al. 2016]). This simulation yields the 'density' of the system in osmotic equilibrium (in fact it yields the box volume V). In a second step, a NVT simulation should be carried out at that 'density'. In this simulation, the osmotic pressure Π is monitored.

Additionally, the OPAS version of the $ms2$ program computes density profiles for all components. These are required to calculate the composition of the solution phase, as it fluctuates during the simulation. However, these fluctuations are usually very small once equilibrium is attained. The OPAS version of $ms2$ outputs an additional file with extension *.dcp. It contains the number density of all components versus the coordinate normal to the membrane. The number of bins for the density profile can be specified, see below.

To obtain a $ms2$ version for performing OPAS simulations, the flag 'OSMOP = 1' must be included when compiling the source code. To assign whether a component can permeate the membrane, the keywords 'Permeability = On' or 'Permeability = Off' must be provided in the *.par file. Additionally, the number of bins for the density profile must be assigned in the *.par file using the keyword 'NumDenBins = 120'. For OPAS simulations, it is recommended to use at least 4000 particles, mainly to obtain unperturbed density profiles.

5.7.3 Multiensembles for multiple state points

In *ms*2, multiple ensembles can be simulated currently with one program execution of *ms*2. This essentially means that one *.`par` file specifies multiple state points at different temperatures or pressures. This can be very convenient, e. g. when different state points have to be simulated with the same simulation setup. Another use case are mixtures, where several considered concentration can be defined in several ensembles of one single simulation run.

To use this feature the number of ensembles has to be defined in the *.`par` file. The specific parameters in the *.`par` file have to be specified once at the beginning followed by the ensemble specific parameters for each ensemble, e. g. state point. To create a reasonable simulation run the individual ensembles should be similar to each other regarding system size, cutoff radius, etc. to prevent strongly different simulation times among the ensembles. Moreover the user should choose an appropriate number of cores that is divisible by the number of ensembles to ensure a technically unproblematic simulation. All simulation output files will be written per ensemble indicated by the ensemble number in the file name.

6 Potential models in the *ms*2 - distribution

The distribution of *ms*2 comes with several molecular models in the *.pm file format. The zip-file containing these force fields is downloaded from the website of the database of "*Molecular Models of the Boltzmann-Zuse Society*" which is also developed and maintained by our group. The models which are documented there are rigid classical mechanical force fields for low-molecular fluids, consisting of 12-6 Lennard-Jones interaction sites, point charges, dipoles and quadrupoles. The database contains force fields for around 150 substances for low-molecular fluids. They are known to describe vapour-liquid equilibrium data (saturated densities, vapour pressure, enthalpy of vaporization) well, to which they were parametrised in most cases. In many cases, also predictions of other properties, like transport properties were tested and found to be in good agreement with experimental data. The following list gives an overview of the substances that are available in that database.

Filename	CAS number	Name	Publication	Model Type
Ar.pm	7440-37-1	Argon	[Vrabec et al. 2001]	1 L.J. site
Ba2+.pm	22541-12-4	Barium cation (2+)	[Deublein et al. 2012]	1 L.J. site & 1 charge
Be2+.pm	22537-20-8	Beryllium ion (2+)	[Deublein et al. 2012]	1 L.J. site & 1 charge
Br-.pm	24959-67-9	Bromine ion (1−)	[Reiser et al. 2014]	1 L.J. site & 1 charge
Br2.pm	7726-95-6	Bromine	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
C2Br2F4.pm	124-73-2	1,2-Dibromotetrafluoroethane	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole
C2Cl4.pm	127-18-4	Tetrachloroethylene	[Stoll 2004]	2 L.J. sites & 1 quadrupole
C2ClF3.pm	79-38-9	Chlorotrifluoroethene	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2F4.pm	116-14-3	Tetrafluoroethylene	[Stoll 2004]	2 L.J. sites & 1 quadrupole
C2F6.pm	76-16-4	Perfluoroethane	[Stoll 2004]	2 L.J. sites & 1 quadrupole
C2H2.pm	74-86-2	Acetylene	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
C2H2Cl3F.pm	27154-33-2	Ethane, trichlorofluoro-	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H2Cl3F.pm	27154-33-2	Ethane, trichlorofluoro-	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H2Cl4_I.pm	630-20-6	1,1,1,2-Tetrachloroethane	[Stoll 2004]	2 L.J. sites & 1 dipole
C2H2Cl4_II.pm	630-20-6	1,1,1,2-Tetrachloroethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H2F2.pm	75-38-7	1,1-Difluoroethene	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H3Cl.pm	75-01-4	Vinyl chloride	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H3Cl3_a.pm	71-55-6	Methylchloroform	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H3Cl3_b.pm	79-00-5	1,1,2-Trichloroethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H3F.pm	75-02-5	Vinyl fluoride	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H3N.pm	75-05-8	Acetonitrile	[Eckl et al. 2008c]	3 L.J. sites & 1 dipole & 1 quadrupole
C2H3N_m6.pm	75-05-8	Acetonitrile	[Deublein et al. 2013]	3 L.J. sites & 1 dipole
C2H3N_m8.pm	75-05-8	Acetonitrile	[Deublein et al. 2013]	3 L.J. sites & 1 dipole
C2H4.pm	74-85-1	Ethylene	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
C2H4Br2.pm	106-93-4	Ethylene dibromide	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole
C2H4Br3.pm	557-91-5	1,1-Dibromoethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H4Cl2.pm	75-34-3	1,1-Dichloroethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H4O_I.pm	75-21-8	Ethylene oxide	[Eckl et al. 2008b]	3 L.J. sites & 1 dipole
C2H5Br.pm	74-96-4	Ethyl bromide	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H5F.pm	353-36-6	Fluoroethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
C2H6_I.pm	74-84-0	Ethane	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
C2H6_II.pm	74-84-0	Ethane	[Stoll 2004]	2 L.J. sites & 1 quadrupole
C2H6O.pm	64-17-5	Ethanol	[Schnabel et al. 2005]	3 L.J. sites & 3 charges
C2H6O2.pm	107-21-1	Ethylene glycol	[Huang et al. 2012]	4 L.J. sites & 6 charges
C2H6S_I.pm	75-18-3	Dimethyl sulfide	[Eckl et al. 2008c]	3 L.J. sites & 1 dipole & 2 quadrupoles
C2H8N2.pm	57-14-7	1,1-Dimethylhydrazine	[Elts et al. 2012]	4 L.J. sites & 3 charges
C2HCl3.pm	79-01-6	Trichloroethylene	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole

Continued on next page

Filename	CAS number	Name	Publication	Model Type
C2N2.pm	460-19-5	Cyanogen	[Miroshnichenko et al. 2013]	4 L.J. sites & 1 quadrupole
C3H4.pm	463-49-0	Propadiene	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
C3H6_a.pm	115-07-1	Propylene	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
C3H6_b.pm	75-19-4	Cyclopropane	[Munoz-Munoz et al. 2015]	3 L.J. sites
C4F10.pm	355-25-9	Perfluorobutane	[Köster et al. 2012]	14 L.J. sites & 14 charges
C4H10.pm	75-28-5	Isobutane	[Eckl et al. 2008c]	4 L.J. sites & 1 dipole & 1 quadrupole
C4H4S_I.pm	110-02-1	Thiophene	[Eckl et al. 2008c]	5 L.J. sites & 1 dipole & 1 quadrupole
C4H8.pm	287-23-0	Cyclobutane	[Munoz-Munoz et al. 2015]	4 L.J. sites
C5H10_dfg.pm	287-92-3	Cyclopentane	[Munoz-Munoz et al. 2015]	5 L.J. sites
C5H10_diff.pm	287-92-3	Cyclopentane	[Munoz-Munoz et al. 2015]	5 L.J. sites
C6H10O.pm	108-94-1	Cyclohexanone	[Merker et al. 2012]	7 L.J. sites & 1 dipole
C6H12_dfg.pm	110-82-7	Cyclohexane	[Munoz-Munoz et al. 2015]	6 L.J. sites
C6H12_diff.pm	110-82-7	Cyclohexane	[Munoz-Munoz et al. 2015]	6 L.J. sites
C6H12_I.pm	110-82-7	Cyclohexane	[Eckl et al. 2008c]	6 L.J. sites & 1 quadrupole
C6H12_II.pm	110-82-7	Cyclohexane	[Merker et al. 2012]	6 L.J. sites
C6H12O_II.pm	108-93-0	Cyclohexanol	[Merker et al. 2009]	7 L.J. sites & 3 charges 1 quadrupole
C6H4Cl2.pm	95-50-1	ortho-Dichlorobenzene	[Huang et al. 2011]	8 L.J. sites & 1 dipole & 4 quadrupoles
C6H5Cl.pm	108-90-7	Chlorobenzene	[Huang et al. 2011]	7 L.J. sites & 1 dipole & 5 quadrupoles
C6H6_ I.pm	71-43-2	Benzene	[Guevara-Carrion et al. 2016]	6 L.J. sites & 6 quadrupoles
C6H6_II.pm	71-43-2	Benzene	[Huang et al. 2011]	6 L.J. sites & 6 quadrupoles
C7H8_I.pm	108-88-3	Toluene	[Guevara-Carrion et al. 2016]	7 L.J. sites & 6 quadrupoles
C7H8_II.pm	108-88-3	Toluene	[Huang et al. 2011]	7 L.J. sites & 1 dipole & 5 quadrupoles
Ca2+.pm	17787-72-3	Calcium ion (2+)	[Deublein et al. 2012]	1 L.J. site & 1 charge
CBr2F2.pm	75-61-6	Dibromodifluoromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CBrCl3.pm	75-62-7	Bromotrichloromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CBrClF2.pm	353-59-3	Bromochlorodifluoromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CC2HBrClF3.pm	151-67-7	Halothane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CCl2O.pm	75-44-5	Phosgene	[Huang et al. 2011]	4 L.J. sites & 1 dipole & 1 quadrupole
CCl4_I.pm	56-23-5	Carbon tetrachloride	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
CCl4_II.pm	56-23-5	Carbon tetrachloride	[Guevara-Carrion et al. 2016]	5 L.J. sites & 5 charges
CClN.pm	506-77-4	Cyanogen chloride	[Miroshnichenko et al. 2013]	3 L.J. sites & 1 quadrupole & 1 dipole
CF2CFBr.pm	598-73-2	Bromotrifluoroethylene	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CF4_II.pm	75-73-0	Tetrafluoromethane	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
CH2Br2_D.pm	74-95-3	Dibromomethane	[Stoll et al. 2003]	1 L.J. site & 1 dipole
CH2Br2_Q.pm	74-95-3	Dibromomethane	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole
CH2BrCl.pm	74-97-5	Bromochloromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CH2Cl2.pm	75-09-2	Methylene chloride	[Stoll et al. 2003]	1 L.J. site & 1 dipole

Continued on next page

Filename	CAS number	Name	Publication	Model Type
CH2I2.pm	75-11-6	Methylene iodide	[Stoll et al. 2003]	1 L.J. site & 1 dipole
CH2O.pm	50-00-0	Formaldehyde	[Eckl et al. 2008c]	2 L.J. sites & 1 dipole
CH2O2.pm	64-18-6	Formic acid	[Schnabel et al. 2007b]	3 L.J. sites & 4 charges
CH3Br.pm	74-83-9	Methyl bromide	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CH3I.pm	74-88-4	Methyl iodide	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CH3NO2.pm	75-52-5	Nitromethane	[Eckl et al. 2008c]	4 L.J. sites & 1 dipole & 1 quadrupole
CH3OCH3.pm	115-10-6	Dimethyl ether	[Eckl et al. 2008c]	3 L.J. sites & 1 dipole
CH4.pm	74-82-8	Methane	[Vrabec et al. 2001]	1 L.J. site
CH4O_I.pm	67-56-1	Methyl alcohol	[Schnabel et al. 2007a]	2 L.J. sites & 3 charges
CH5N.pm	74-89-5	Methylamine	[Schnabel et al. 2008]	2 L.J. sites & 2 charges
CH6N2.pm	60-34-4	Methylhydrazine	[Elts et al. 2012]	3 L.J. sites & 3 charges
CHBr3.pm	75-25-2	Bromoform	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CHCCH3_I.pm	74-99-7	1-Propyne	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
CHCCH3_II.pm	74-99-7	1-Propyne	[Stoll 2004]	2 L.J. sites & 1 quadrupole
CHCl2F.pm	75-43-4	Dichlorofluoromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CHCl3.pm	67-66-3	Chloroform	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CHN.pm	74-90-8	Hydrogen cyanide	[Eckl et al. 2008c]	2 L.J. sites & 1 dipole & 1 quadrupole
Cl-.pm	16887-00-6	Chlorine ion (1−)	[Reiser et al. 2014]	1 L.J. site & 1 charge
Cl2.pm	7782-50-5	Chlorine	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
CO_D.pm	630-08-0	Carbon monoxide	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
CO_Q.pm	630-08-0	Carbon monoxide	[Stoll 2004]	2 L.J. sites & 1 quadrupole
CO2_I.pm	124-38-9	Carbon dioxide	[Merker et al. 2010]	3 L.J. sites & 1 Quattropole
CO2_II.pm	124-38-9	Carbon dioxide	[Stoll 2004]	2 L.J. sites & 1 quadrupole
Cs+.pm	18459-37-5	Cesium ion (1+)	[Reiser et al. 2014]	1 L.J. site & 1 charge
CS2.pm	75-15-0	Carbon disulfide	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
F-.pm	16984-48-8	Fluorine ion (1−)	[Reiser et al. 2014]	1 L.J. site & 1 charge
F2.pm	7782-41-4	Fluorine	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
H3N_II.pm	7664-41-7	Ammonia	[Eckl et al. 2008a]	1 L.J. site & 4 charges
H4N2.pm	302-01-2	Hydrazine	[Elts et al. 2012]	2 L.J. sites & 6 charges
HCl.pm	7647-01-0	Hydrochloric acid	[Huang et al. 2011]	1 L.J. site & 2 charges
I-.pm	20461-54-5	Iodide ion (1−)	[Reiser et al. 2014]	1 L.J. site & 1 charge
I2.pm	7553-56-2	Iodine	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
K+.pm	24203-36-9	Potassium ion (1+)	[Reiser et al. 2014]	1 L.J. site & 1 charge
Kr.pm	7439-90-9	Krypton	[Vrabec et al. 2001]	1 L.J. site
Li+.pm	17341-24-1	Lithium ion (1+)	[Reiser et al. 2014]	1 L.J. site & 1 charge
Mg2+.pm	22537-22-0	Magnesium ion (2+)	[Deublein et al. 2012]	1 L.J. site & 1 charge
N2.pm	7727-37-9	Nitrogen	[Stoll 2004]	2 L.J. sites & 1 quadrupole

Continued on next page

Filename	CAS number	Name	Publication	Model Type
Na+.pm	17341-25-2	Sodium ion (1+)	[Reiser et al. 2014]	1 L.J. site & 1 charge
Ne.pm	7440-01-9	Neon	[Vrabec et al. 2001]	1 L.J. site
O2.pm	7782-44-7	Oxygen	[Stoll 2004]	2 L.J. sites & 1 quadrupole
R11_CFCl3.pm	75-69-4	Trichloromonofluoromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R1122_CHCl=CF2.pm	359-10-4	2-Chloro-1,1-difluoroethylene	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R112a_CCl3-CF2Cl.pm	76-11-9	1,1,1,2-Tetrachloro-2,2-difluoroethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R113_CFCl2-CF2Cl.pm	76-13-1	1,1,2-trichloro-1,2,2-trifluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole
R114_CF2Cl-CF2Cl.pm	76-14-2	1,2-dichloro-1,1,2,2-tetrafluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole
R115_CF3-CF2Cl.pm	76-15-3	Pentafluoroethyl chloride	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole
R12_CF2Cl2.pm	75-71-8	Dichlorodifluoromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R123_CHCl2-CF3.pm	306-83-2	2,2-dichloro-1,1,1-trifluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R124_CHFCl-CF3.pm	2837-89-0	2-chloro-1,1,1,2-tetrafluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R125_CHF2-CF3.pm	354-33-6	pentafluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R13_CF3Cl.pm	75-72-9	Chlorotrifluoromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R134_CHF2-CHF2.pm	359-35-3	1,1,2,2-tetrafluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 quadrupole
R134a_CH2F-CF3.pm	811-97-2	Norflurane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R13B1_CBrF3.pm	75-63-8	Bromotrifluoromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R141b_CH3-CFCl2.pm	1717-00-6	1,1-Dichloro-1-fluoroethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R142b_CH3-CF2Cl.pm	75-68-3	1-chloro-1,1-difluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R143a_CH3-CF3.pm	420-46-2	1,1,1-trifluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R152a_CH3-CHF2.pm	75-37-6	1,1-difluoro-Ethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R22_CHF2Cl.pm	75-45-6	Difluorochloromethane	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R227ea_C3HF7.pm	431-89-0	1,1,1,2,3,3,3-heptafluoro-Propane	[Eckl et al. 2007]	10 L.J. sites & 1 dipole 1 quadrupole
R23_CHF3.pm	75-46-7	Fluoroform	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
R32_CH2F2.pm	75-10-5	Difluoromethane	[Stoll et al. 2003]	1 L.J. site & 1 dipole
R41_CH3F.pm	593-53-3	Methyl fluoride	[Stoll et al. 2003]	2 L.J. sites & 1 dipole
Rb+.pm	22537-38-8	Rubidium ion (1+)	[Reiser et al. 2014]	1 L.J. site & 1 charge
SF6.pm	2551-62-4	Sulfur hexafluoride	[Vrabec et al. 2001]	2 L.J. sites & 1 quadrupole
SO2.pm	7446-09-5	Sulfur dioxide	[Eckl et al. 2008c]	3 L.J. sites & 1 dipole & 1 quadrupole
Sr2+.pm	22537-39-9	Strontium ion (2+)	[Deublein et al. 2012]	1 L.J. site & 1 charge
Xe.pm	7440-63-3	Xenon	[Vrabec et al. 2001]	1 L.J. site

7 Computational aspects

7.1 Parallelization

The source code is developed for an efficient parallelism with distributed memory, using the PI standard for communication [Forum 2009]. The program uses the following MPI calls:

- MPI_Init, MPI_Finalize, MPI_Comm_rank, MPI_Comm_size to set up basic MPI functionality
- MPI_Abort to stop the program in case of an error
- MPI_Barrier, MPI_Wtime and MPI_Wtick within the stopwatch class
- MPI_Bcast, MPI_Reduce, MPI_Allreduce for simulation data exchange

Exclusively collective communication is used, an approach that is suitable for molecular simulations, where the cut-off radius is in the order of the simulation volume edge length. For MD simulations with $ms2$, only the force calculation is parallelized using the molecule decomposition according to Plimpton [Plimpton 1993]. The interaction matrix is rearranged such that the number of interaction partners is almost equally distributed in the matrix. This is achieved by calculating the interactions between molecule 1 and N as interactions between molecules N and 1, cf. Figure 7.1. The parallelization scheme then distributes the calculation among N_P processors with an almost equal work load such that every processor executes N/N_P rows of the interaction matrix. The master process reduces then all the force components to sum up the resulting molecular forces on each molecule.

The MC code is parallelized exploiting the stochastic nature of the simulation method. Starting from an equilibrated configuration in the simulation volume, the molecular configuration is copied multiple times into different volumes. Then, each copy runs independently parallel, but with a different random number seeds, to calculate the desired thermodynamic properties. At the end, the data from all copies are gathered and averaged.

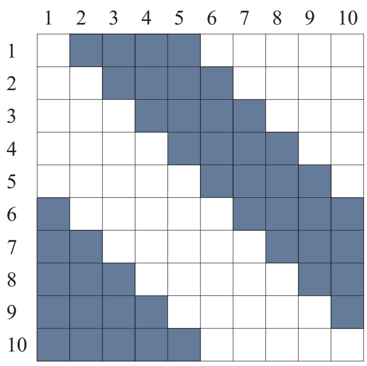


Figure 7.1: Schematic of the molecule decomposition parallelization by Plimpton [Plimpton 1993].

7.2 Vectorization & memory

The program *ms*2 specifically accounts for vector-parallel machines. All data needed in the computationally costly inner loops of the calculation, like the positions of the molecules and their sites, are stored in vectors that are accessed sequentially. This allows for an efficient use of memory on vector-parallel machines.

For the evaluation of one configuration in *ms*2, e. g. the position vectors are loaded once and then distributed among all processors. Subsequently, additional information like interaction partners of each individual molecule is computed, tabulated in vectors and communicated. The order of these data follows the order of all other vectors, e. g. the position vector. An example of a storage vector is shown in Table 7.1. The appropriate order allows for a serial access to the data in all vectors in the random access memory. Therefore, the calculations can be vectorized, increasing efficiency.

Table 7.1: Listing of the interaction partners for a molecule i, indicating the assembly in the storage vector.

Molecule i	Interaction list for molecule i in storage vector				
1	2	4	6	...	10
3		4	7	...	10
4			8	...	10

7.3 Structure of program code

This section discusses the code design and implementation issues of *ms*2 in detail. These explanations of the source code are intended to help understanding the program and to encourage the reader to further develop and improve the code. Due to the experience of the core developers and the suitability for an efficient numerical code, Fortran90 was chosen as programming language. The program makes extensive use of the concepts introduced with Fortran90, notably modular programming.

A wide variety of different simulation setups is possible, each requiring different computations. The most significant difference between the setups lies with the choice of MD or MC. While MD features extensive force calculations in order to update the molecular positions, MC is based on a Markov chain, where a molecular move is accepted with a certain probability. However, there are many other choices regarding the setup. This variety leads to a strong need for a highly modular structure, where modules have clear-cut interfaces. Given such a modularity, the appropriate functionalities can be combined to the desired simulation, while leaving out any unnecessary code[1]. Besides avoiding if-statements at computationally demanding points and therefore leading to a better data flow, this allows for an efficient implementation of new functionalities, as existing modules can be used whenever feasible. In *ms*2, a stringent philosophy regarding modularization was followed. The runs need to be fast and scale well with increasing number of processors in order to achieve short response times. *ms*2 is parallelized with the Message Passing Interface (MPI) [Forum 2009]. While MD needs synchronization after every time step due to its deterministic nature, MC can be run embarrassingly parallel, only requiring synchronization at the very end of a run[2]. Furthermore, the by far most expensive part of the code, calculating forces and interaction energies, is highly vectorized and optimized.

In order to keep the class hierarchy flat, *ms*2 does not make use of inheritance and hence polymorphism. E. g. the module ms2_site contains the classes TSiteLJ126, TSiteCharge, TSiteDipole and TSiteQuadrupole, which have data and functionalities in common, but they are not derived from a common class. The class structure and the relations between the classes is organized in six levels, as depicted in Figure 7.2. Every class is located on a certain level and hence has data and modules relevant to its level. The six levels are intuitive:

1. global – all data and functions of global use are handled. Scope: whole code
2. simulation – setup and control of the simulation. Scope: simulation framework
3. ensemble – the required ensemble is set up and initialized. Scope: molecular ensemble
4. component – data and functions dealing with all molecules of a given molecule type. Scope: one molecule species
5. molecule – data regarding the basic structure of a molecule. Scope: one molecule
6. site – position of sites with respect to a molecule. Scope: site of one molecule

First level - global:
The first level contains subroutines and variables that are needed globally in all parts of the simulation. The variables are mainly natural constants like the Boltzmann constant k_B, the Avogadro constant N_A, etc. Additionally, variables defining the simulation setup for the given run are stored, e. g. the simulation technique and the ensemble type. These variables determine the type of calculation and are assigned according to user specifications on subsequent levels in *ms*2. Subroutines that are stored on this level are also of global use in *ms*2. These are basic routines for input into the simulation tool and output into files, as well as more advanced routines for automated communication between hardware and program via signals. The latter routines are of particular interest for running *ms*2 on high performance clusters, where possible data loss can be avoided by automated communication between cluster nodes and the program.

Second level - simulation:
On the second level, the simulation flow is controlled. The simulation is initiated, started and sustained. This includes controlling the length of the simulation and its output. User specifications concerning the simulation setup are read and the corresponding global variables are assigned accordingly. Furthermore, all simulation quantities and averages are analyzed and written to file.

Third level - ensemble:
On the third level, the simulation is organized according to the desired ensemble specified by the user. The respective ensemble is initialized, by characterizing the ensemble setup and defining e. g. the volume and the composition of the molecular species in the simulation volume. In addition, the initial positions and velocities of all molecules are distributed. Furthermore, the routines for the

[1] Unnecessary for the current setup.
[2] Therefore, parallelization can be achieved by simply executing multiple MC runs simultaneously and averaging over all runs

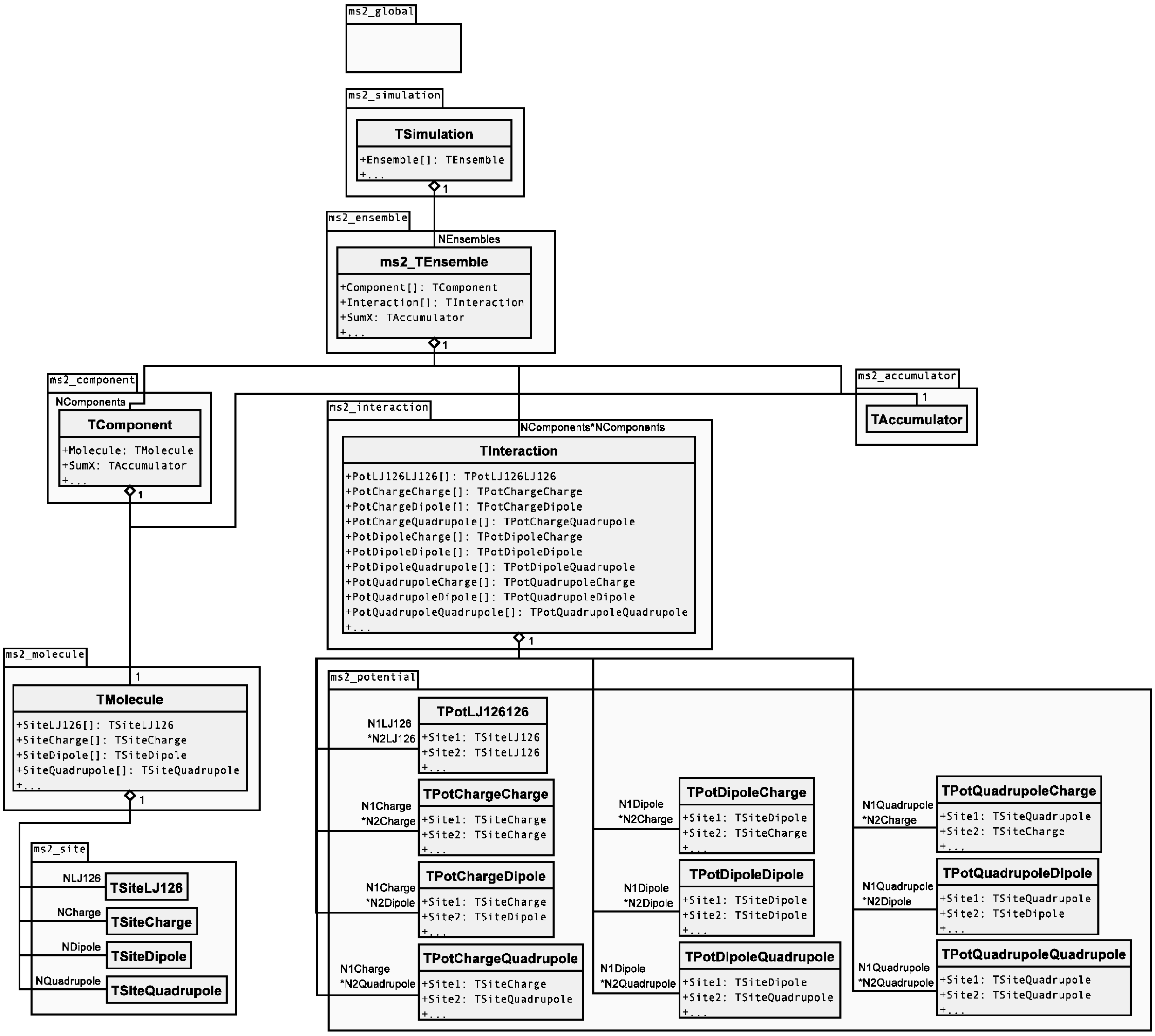


Figure 7.2: UML class diagram of *ms*2, showing object attributes.

subsequent simulation are assigned on this level. The time evolution or the Markov chain, respectively, is performed and the ensemble averages of thermodynamic properties are calculated during the simulation run.

Fourth level - component:
On this level, the molecule species (component) and their interactions are addressed. The class structure has three distinct branches. While the first branch deals with issues regarding structure and properties of one component at a time, the second branch deals with interactions between molecules of one or two components. The third branch contains all routines for the accumulation of data, determining the average value of the data as well as their statistical uncertainties. The first two branches will be discussed in more detail.
The first branch deals with the structure of one individual component. An instance of the class component stores center of mass position, orientation, velocity etc. of every molecule of one species. The contained modules deal with calculations all molecules of this species, e. g. modules for solving Newton's equations of motion, kinetic energy calculation, atom to molecule transformation and vice versa. Furthermore, this branch is responsible for the correct initialization. It reads the molecular model for every component, calculates positions, introduces the reduced quantities, etc.
The second branch deals with interactions between molecules of one or two components. Therefore, this branch contains all force and potential energy calculations, which are the computationally most expensive parts by far. This branch is highly vectorized to achieve a fast calculation, in particular if executed in vector-parallel. The modules in this branch are simulation technique specific, i. e. MD or MC.
In a MD simulation, first, the interaction partners of every molecule are determined, i. e. other molecules or sites within the cut-off radius. Then the interaction energies and virial contributions as well as the resulting forces and torques are calculated and summed up for every molecule.
In a MC simulation, the modules are designed to calculate only the energy and the virial. Forces are not calculated and the corresponding algorithms are therefore entirely omitted in these modules. For MD, all interactions of one component-component pair[3] are calculated in one call of the module and they are therefore called once for every component-component pairing. For MC, the modules are finer-grained, calculating all interactions of one component-component pair by determining every molecule-molecule interaction individually. The corresponding modules are thus called more frequently than in case of MD.

Fifth level - molecule:
The basic information on molecules is stored on this level, such as mass, moment of inertia tensor, rotational degrees of freedom, etc. Higher level modules access this data via the class molecule.

Sixth level - site:
Here, the assembly of a molecule is stored. The data is used by higher level modules to calculate the site positions from the molecular positions and orientations. This allows for the calculation of site-site interactions. It should be noted that $ms2$ only integrates (MD) or accepts (MC) center of mass positions and orientations. Site specific data is calculated for every simulation step on the fly.

Example:
The modular structure is discussed by highlighting the calculation process of a MD simulation of a pure fluid modeled by two LJ sites and two point charges in the NVT ensemble. During the initialization process, the modules for memory allocation and definition of the simulation scenario are executed. This process is not unique for one of the hierarchical levels, but takes places on all levels, cf. Figure 7.2. On level one, the simulation is delimited to "molecular dynamics", therefore all modules concerning MC simulations are neglected entirely. On level two, the "initialization" modules define the simulation environment by setting ensemble specific properties like simulation volume, temperature as well as the thermostat and integration scheme, etc. While the use of the first modules mentioned here is

[3]Please note that a component-component pair can consist of one component paired with itself.

required in each simulation, independently of MD or MC, e. g. the trajectory calculation is specific for MD. However, the modular structure helps keeping the efficiency of the program by initiating only the simulation-specific molecule propagation. The same effect of modules can be found in the initiation process on lower levels. Here, the exact composition of the molecular species in the simulation volume (module component), the description of the molecule (module molecule) and the storage of the positions, velocities and forces of all LJ sites and charges (module sites) is determined. All other modules, like treating additional components for mixtures or containing characteristic properties of dipoles or quadrupoles, are fully omitted by $ms2$, since these are not used throughout this simulation.

Once the simulation has been initialized, a set of modules is invoked, which deals with maintaining the molecular simulation. On level one, this includes modules of global use for any simulation, e. g. limiting the execution time, etc. On level two, Newton's equations of motion are numerically integrated, requiring information about the fluid's composition, the positions and orientations of the molecules etc. provided by modules of the respective level. The thermostat is applied and the calculation of thermodynamic properties is performed. The forces, torques and energies are calculated in the second branch on the level component. For a MD simulation, this covers routines for the calculation of energies as well as forces at the same time, whereas in contrast for a MC simulation, the modules only calculate energies and virial contributions.

A further set of modules deals with the accumulation of data. These modules are designed to store data and evaluate the statistics of the produced data. The actual thermodynamic properties are calculated on the ensemble level. The last set of modules deals with the output, where the results of the molecular simulation are written to file. These modules are called independent on the user settings in all simulation scenarios.

8 Tools

The tools described in the following are intended to facilitate the handling of *ms*2. They allow an easy start of molecular simulations with *ms*2 and give access to the output data generated by *ms*2. The tool *ms2par* is a GUI to facilitate the composing of *.`par` files for different tasks, while *ms2chart* simply plots the simulation results (equilibration and production phase) with its statistical uncertainties. *ms2molecules* is a visualization tool for the displaying of the trajectories obtained by *ms*2. *res2ge_par* is shell script for automatically generating input *.`par` files for Grand Equilibrium ensemble simulations, especially for the investigation of vapor liquid equilibria. All tool run on Linux and Windows platforms and are part of the *ms*2 distribution.

8.1 *ms*2par

The GUI-based feature tool *ms*2*par* allows for a convenient generation of *.`par` files according to user specifications. It assigns the molecular model, ensemble, thermodynamic state point, time step, number of equilibration steps, etc. successively. Furthermore, the tool proposes a cut-off radius based on the input data. After the user completes the specifications, the program generates the file *.`par` in ASCII format, to be read by *ms*2. Applying *ms*2*par* is simple and should be intuitive, also for new users. *ms*2*par* is a java application, thus it runs independently on all operating systems.

8.2 *ms*2chart

*ms*2*chart* is a GUI-based java applet for evaluating the simulation results from *ms*2. This tool plots the evolution of properties calculated by *ms*2. The properties shown on the different graph axes are chosen from a drop down menu and the plot is shown directly in the GUI.
For a better analysis, *ms*2*chart* allows various features: plotting block averages as well as simulation averages into the same plot, changing the design of the plot, individual labeling of the axes and adjusting the frame detail of the plot. All plots can be exported in png format. If needed, the simulation details can be displayed, i. e. the specifications in the *.`par` file, the results in the *.`res` file and the summary in the *.`log` file.
The analysis program *ms*2*chart* can be executed at any time of the simulation, i. e. also while the simulation is still ongoing. There is no loss of data, if it is run on the fly. The tool *ms*2*chart* is easy to understand and can be handled intuitively. It allows for an easy first analysis of the *ms*2 simulation results. Advanced users will use *ms*2*chart* for a quick check of their simulation results, such as whether the equilibration time was appropriate or for a fast analysis, which is important if extensive series of simulation runs are performed.

8.3 *ms*2molecules

The program *ms*2*molecules* is a visualization tool for *ms*2. The program displays the molecular trajectories that are stored by *ms*2 as a series of configurations in the *.`vim` file. *ms*2*molecules* visualizes molecular sites by colored spheres. The colors are user defined. The size of a sphere is by default proportional to the LJ size parameter σ, but can be changed manually in the first lines of the

*.vim file. This feature facilitates monitoring a component of interest by reducing the size of other components, e. g. solvents. Other features like zooming in or out of the simulation volume and rotating the simulation box also facilitates the analysis of the simulation. The visualization can be exported via snapshots in the jpg format. The program is based on OpenGL and written in C. The handling of the program is simple and console based. Requirements for the feature tool are OpenGL in a Windows or Linux environment.

Colors:
Setting the colors of the molecules in the visualization window has to be done manually in the header of the *.vim file. The particle color is defined by the last integer value of each line in the header, which makes the visualization easy adjustable. The color of each cite-type of each molecule-type in the simulation can be set separately. The available colors are:

Table 8.1: Available colors in *molecules*

Integer in vim file	color
1	black
2	red
3	orange
4	yellow
5	yellow 2
6	green
7	green 2
8	turquoise
9	blue
10	blue 2
11	purple
12	pink
13	pink 2
14	grey

Adjusting Atom size:
The atom size, or the size of one cite in a molecule-type can be adjusted manually by the user in the header of the *.vim file. That can be done by setting the σ value of each site, e. g.

```
~  1 LJ  0.0000 -0.3387 -1.4713  3.0  1
~  1 LJ -0.0000  0.8836  0.0917  1.42  2
~  1 LJ -0.0000 -0.4503  1.1711  1.8  3
```

The integer "1" in the first column after the tilde, specifies the LJ-sites as one of the 1st molecule (used *.pm file). The 2nd, 3rd and 4th numeric column are the coordinates of the site relative to the center of mass of the molecule. The 5th column indicates the σ value of the site and can manually re-adjusted. The last integer column indicates the color of the site (see Tab. 8.1 above).

8.4 res2gepar

res2ge_par is a bash-script that prepares an input *.par file for a following simulation using the Grand Equilibrium ensemble. It requires the data of a prior simulation in the NpT ensemble as input (Sim1). The syntax to type in the command line is as follows:

```
res2gepar Sim1.par Sim1_1.res > ge_run.par
```

The program `res2gepar` needs to be copied to the current directory where also `Sim1.par` and `Sim1_1.res` are saved. When `res2gepar` is executed, it expects two arguments: the *`.par` file and the *`.res` file from the liquid phase simulation. The new `ge_run.par`

file is the input file for the calculation of the coexisting phase of Sim1. Its name can be set arbitrarily by the user.

Bibliography

Alfe, D. and M. J. Gillan (1988). "First-Principles Calculation of Transport Coefficients". In: *Phys. Rev. Lett.* 81, pp. 5161–5164 (cit. on p. 28).

Allen M.P. & Tilldesley, D.J. (2002). *Computer Simulations of Liquids*. Oxford: Oxford University Press (cit. on pp. 17–21, 23, 26, 27).

Arunan, E, G.R. Desirajiu, R.A. Klein, J. Sadlej, S. Scheiner, and I. Alkorta (2011). "Demodels the hydrogen bond: An account (IUPAC technical report)". In: *Pure and Applied Chemistry* 83.8, pp. 1619–1636. DOI: 10.1351/PAC-REP-10-01-01 (cit. on p. 30).

Barker, J. A. and R. O. Watts (1973). In: *Mol. Phys.* 26, pp. 789–792 (cit. on p. 26).

Berthelot, D. (1898). In: *Comptes Rendues Hebdomadaires des Seances de l'Academie des Sciences* 126, p. 1703 (cit. on p. 22).

Boettcher C., van Belle O. Bordewijk P. and A. Rip (1973). *Theory of electric polarization, Vol. 1: Dielectrics in static fields*. Elsevier (cit. on p. 23).

Born, M. and T. von Karman (1912). In: *Physikalische Zeitschrift* 13, p. 297 (cit. on p. 17).

Deublein, S., S. Reiser, J. Vrabec, and H. Hasse (2012). "A Set of Molecular Models for Alkaline-Earth Cations in Aqueous Solution". In: *Journal of Physical Chemistry B* 116.18, p. 5448. DOI: 10.1021/jp3013514 (cit. on pp. 39–42).

Deublein, S., P. Metzler, J. Vrabec, and H. Hasse (2013). "Automated development of force fields for the calculation of thermodynamic properties acetonitrile as a case study". In: *Molecular Simulation* 39.2, p. 109. DOI: 10.1080/08927022.2012.705434 (cit. on p. 39).

Eckl, B., Y. Huang, and H. Vrabec J.and Hasse (2007). "Vapor pressure of R227ea + ethanol at 343.13K by molecular simulation". In: *Fluid Phase Equilibria* 260, 177–182. DOI: 10.1016/j.fluid.2007.05.011 (cit. on p. 42).

Eckl, B., J. Vrabec, and H. Hasse (2008a). "An optimised molecular model for ammonia". In: *Molecular Physics: An International Journal at the Interface Between Chemistry and Physics* 106.8, p. 1039. DOI: 10.1080/00268970802112137 (cit. on p. 41).

Eckl, B., J. Vrabec, and H. Hasse (2008b). "On the application of force fields for predicting a wide variety of properties: Ethylene oxide as an example". In: *Fluid Phase Equilibria* 274.1. DOI: 10.1016/j.fluid.2008.02.002 (cit. on p. 39).

Eckl, B., J. Vrabec, and H. Hasse (2008c). "Set of Molecular Models Based on Quantum Mechanical Ab Initio Calculations and Thermodynamic Data". In: *The Journal of Physical Chemistry B* (cit. on pp. 39–42).

Elts, E., T. Windmann, D. Staak, and J. Vrabec (2012). “Fluid phase behavior from molecular simulation Hydrazine, Monomethylhydrazine, Dimethylhydrazine and binary mixtures containing these compounds”. In: *Fluid Phase Equilibria* 322. DOI: 10.1016/j.fluid.2012.03.008 (cit. on pp. 39, 41).

Evans, D.J. and W. B. Streett (1978). “Transport Properties of Homonuclear Diatomics. 2. Dense Fluids”. In: *Molecular Physics* 36.1, pp. 161–176 (cit. on p. 29).

Ewald, P. (1921). In: *Annalen der Physik* 64, p. 253 (cit. on p. 22).

Fasnacht, M, R.H. Swendsen, and J.M. Rosenberg (2004). “Adaptive integration method for Monte Carlo simulations”. In: *Physical Review E* 69, p. 056704 (cit. on p. 34).

Fischer J., Müller D. Chialvo A. and J. M. Halle (1989). “The influence of unlike molecule interaction parameters on liquid mixture excess properties”. In: *Fluid Phase Equilibria* (cit. on p. 22).

fluidproperties (2010). Industrial fluid properties simulation collective. URL: http://www.fluidproperties.org (cit. on p. 1).

Flyvberg, H. and H. Petersen (1989). In: *The Journal of Chemical Physics* 91, p. 461 (cit. on p. 25).

Forum, Message Passing Interface (2009). *MPI: A Message-Passing Interface Standard, Version 2.2.* High Performance Computing Center Stuttgart (HLRS) (cit. on pp. 43, 45).

Frenkel Daan & Smit, Berend (2002). *Understanding Molecular Dynamics: From Algorithms to Applications.* 2nd edition. San Diego: Academic Press (cit. on pp. 17, 20, 22, 23, 34).

Garzon B., Lago S. Vega C. and L. Rull (1995). In: *The Journal of Chemical Physics* 102, p. 7204 (cit. on p. 22).

Gray, C. and K. Gubbins (1984). *Theory of molecular fluids, Volume 1: Fundamentals.* Oxford: Clarendon Press (cit. on pp. 20–22, 33).

Gubbins, K.E. (1972). *Statistical Mechanics.* Vol. 1. London: The Chemical Society Burlington House (cit. on pp. 28, 29).

Guevara-Carrion, G., T. Janzen, Y. Munoz-Munoz, and J. Vrabec (2016). “Mutual diffusion of binary liquid mixtures containing methanol, ethanol, acetone, benzene, cyclohexane, toluene, and carbon tetrachloride”. In: *The Journal of Chemical Physics* 144. DOI: 10.1063/1.4943395. URL: http://dx.doi.org/10.1063/1.4943395 (cit. on p. 40).

Hansen, J. P. and I. R. McDonald (1986). *Theory of Simple Liquids.* Academic, New York (cit. on p. 29).

Haughney, M., M. Ferrario, and I.R. McDonald (1987). “Molecular-Dynamics Simulation of liquid Methanol”. In: *The Journal of Physical Chemistry* 91.19, p. 4934. DOI: 10.1021/j100303a011 (cit. on pp. 31, 32).

Hirschfelder J., Curtiss C. and R. Bird (1954). *Molecular Theory of Gases and Liquids.* J. Wiley & Sons (cit. on pp. 21, 22).

Hoheisel, C. (1994). "Transport properties of molecular liquids". In: *Physics Reports* 245.3, pp. 111–157. URL: http://www.sciencedirect.com/science/article/B6TVP-46SXPNP-7S/1/41a3ddc287aafd50108d19116a0d6872 (cit. on p. 28).

Huang, Y., M. Heilig, J. Vrabec, and H. Hasse (2011). "Vapor-Liquid Equilibria of Hydrogen Chloride, Phosgene, Benzene, Chlorobenzene, Ortho-Dichlorobenzene and Toluene by Molecular Simulation". In: *AIChE Journal* 57.4, p. 1043. DOI: 10.1002/aic.12329 (cit. on pp. 40, 41).

Huang Y. aHuang, Y. T., M. Heilig, H. Hasse, and J. Vrabec (2012). "Molecular Modeling and Simulation of Vapor–Liquid Equilibria of Ethylene Oxide, Ethylene Glycol, and Water as Well as their Binary Mixtures". In: *Ind. Eng. Chem. Res.* 51.21. DOI: 10.1021/ie300248z (cit. on p. 39).

Impey, R.W., M.A. Madden, and I.R. McDonald (1983). In: *J. Phys. Chem.* 87, p. 5071 (cit. on p. 30).

Jedlovszky, P. and M. Mezei (1999). In: *The Journal of Chemical Physics* 110, p. 2991 (cit. on p. 22).

Kohns, M., S. Reiser, M. Horsch, and H. Hasse (2016). "Solvent activity in electrolyte ssolution from molecular simulation of the osmotic pressure". In: *J. Chem. Phys.* 144, p. 084112. DOI: 10.1063/1.4942500 (cit. on p. 36).

Krishna, R. and J. M. van Baten (2005). "The Darken Relation for Multicomponent Diffusion in Liquid Mixtures of Linear Alkanes: An Investigation Using Molecular Dynamics (MD) Simulations". In: *Ind. Eng. Chem. Res.* 44, pp. 6939–6847 (cit. on p. 28).

Köster, A., P. Nandi, T. Windmann, D. Ramjugernath, and J. Vrabec (2012). "Vapor–liquid equilibria of ethylene (C2H4) + decafluorobutane (C4F10) at 268–298 K from experiment, molecular simulation and the Peng–Robinson equation of state". In: *Fluid Phase Equilibria* 336. DOI: 10.1016/j.fluid.2012.08.023 (cit. on p. 40).

Langenbach, K., C. Engin, S. Reiser, M. Horsch, and H. Hasse (2015). "On the simultaneous description of h-bonding and dipolar interactions with point charges in force field models". In: *AIChE Journal* 61.9, p. 2926. DOI: 10.1002/aic.14820 (cit. on p. 30).

Lennard-Jones, J. (1931). "Cohesion". In: *Proceedings of the Physical Society* 43.5, p. 461 (cit. on p. 20).

Lisal M., Budinsky R. and V. Vacek (1997). In: *Fluid Phase Equilibria* 135, p. 193 (cit. on p. 22).

Lisal M., William W. and I. Nezbeda (2001). In: *Fluid Phase Equilibria* 181, p. 127 (cit. on p. 22).

Lorentz, H. (1881). In: *Annalen der Physik* 127 (cit. on p. 22).

Lorrain P., Corson D. and F. Lorrain (1995). *Elektromagnetische Felder und Wellen.* Berlin: Walter de Gruyter (cit. on p. 21).

Lustig, R. (1994). In: *J. Chem. Phys.* 100, pp. 3060–3067 (cit. on p. 26).

Lustig, R. (1998). In: *Mol. Phys.* 65, pp. 175–179 (cit. on p. 26).

Lustig, R. (2011). In: *Mol. Sim.* 37, pp. 457–465 (cit. on pp. 25, 26).

Lustig, R. (2012). In: *Mol. Phys.* 110, pp. 3041–3052 (cit. on pp. 25, 26).

Lyubartsev, A. P. and A. Laaksonen (2000). "M.DynaMix – a scalable portable parallel MD simulation package for arbitrary molecular mixtures". In: *Computer Physics Communications* 128, p. 565 (cit. on p. 1).

Lyubartsev A. P., Martinovski A. A. Shevkunov S. V. and P. N. Vorontsov-Velyaminov (1992). In: *The Journal of Chemical Physics* 96, p. 1776 (cit. on p. 35).

Meier, K. and S.n Kabelac (2006). In: *J. Chem. Phys.* 124, p. 064104 (cit. on p. 26).

Merker, T., J. Vrabec, and H. Hasse (2012). "Molecular simulation study on the solubility of carbon dioxide in mixtures of cyclohexane + cyclohexanone". In: *Fluid Phase Equilibria* 315, p. 77. DOI: 10.1016/j.fluid.2011.11.003 (cit. on p. 40).

Merker, T., G. Guevara-Carrion, J. Vrabec, and H. Hasse (2009). *HIGH PERFORMANCE COMPUTING IN SCIENCE AND ENGINEERING '08*. Ed. by W. E. Nagel, D. B. Kroner, and M. M. Resch. Springer. DOI: 10.1007/978-3-540-88303-6_37 (cit. on p. 40).

Merker, T., C. Engin, J. Vrabec, and H. Hasse (2010). "Molecular model for carbon dioxide optimized to vapor-liquid equilibria". In: *The Journal of Chemical Physics* 132, p. 234512. DOI: 10.1063/1.3434530 (cit. on p. 41).

Metropolis N., Rosenbluth A. Rosenbluth M. N. Teller A. H. and E. Teller (1953). In: *The Journal of Chemical Physics* 21, p. 1087 (cit. on p. 17).

Miroshnichenko, S., T. Grützner, D. Staak, and J. Vrabec (2013). "Molecular simulation of the vapor–liquid phase behavior of cyanides and their binary mixtures". In: *Fluid Phase Equilibria* 354. DOI: 10.1016/j.fluid.2013.06.039 (cit. on p. 40).

Munoz-Munoz, Y.M., G. Guevara-Carrion, M. Llano-Restrepo, and J. Vrabec (2015). "Lennard-Jones force field parameters for cyclic alkanes from cyclopropane to cyclohexane". In: *Fluid Phase Equilibria* 404, p. 150. DOI: 10.1016/j.fluid.2015.06.033 (cit. on p. 40).

Nezbeda, I. and J. Kolafa (1991). In: *Molecular Simulation* 5, p. 391 (cit. on p. 35).

Omelyan, I. P. (1996). In: *Phys. Lett. A* 223, pp. 295–302 (cit. on p. 26).

Plimpton, S. (1993). "Fast Parallel Algorithms for Short-Range Molecular Dynamics". In: *The Journal of Computational Physics* 117.1, pp. 1–19 (cit. on pp. 1, 43, 44).

Rappaport, D.C. (2004). *The Art of Molecular Dynamics*. 2nd. Cambridge: Cambridge University Press (cit. on pp. 17, 20).

Reiser, S., Deubein, J. S. Vrabec, and H. Hasse (2014). "Molecular dispersion energy parameters for alkali and halide ions in aqueous solution". In: *The Journal of Chemical Physics* 140. DOI: http://dx.doi.org/10.1063/1.4858392 (cit. on pp. 39, 41, 42).

Rowley, R. et al. (2003). "DIPPR Data Compilation of Pure Compound Properties". In: *Design Institute for Physical Properties, AIChE* (cit. on pp. 17, 27).

Schnabel, T., J. Vrabec, and H. Hasse (2005). "Henry's law constants of methane, nitrogen, oxygen and carbon dioxide in ethanol from 273 to 498 K: Prediction from molecular simulation". In: *Fluid Phase Equilibria* 233, pp. 134–143. DOI: 10.1016/j.fluid.2005.04.016 (cit. on p. 39).

Schnabel, T., J. Vrabec, and H. Hasse (2008). “Molecular simulation study of hydrogen bonding mixtures and new molecular models for mono- and dimethylamine”. In: *Fluid Phase Equilibria* 263, 144–159. DOI: `10.1016/j.fluid.2007.10.003` (cit. on p. 41).

Schnabel, T., A. Srivastava, J. Vrabec, and H. Hasse (2007a). “Hydrogen Bonding of Methanol in Supercritical CO2: Comparison between 1H NMR Spectroscopic Data and Molecular Simulation Results”. In: *J.Phys.Chem. B* 111.33, p. 9871. DOI: `10.1021/jp0720338` (cit. on pp. 9, 41).

Schnabel, T., M. Cortada, J. S. Vrabec JVrabec, and H. Hasse (2007b). “Molecular model for formic acid adjusted to vapor–liquid equilibria”. In: *Chemical Physics Letters* 435.4, p. 268. DOI: `10.1016/j.cplett.2006.12.091` (cit. on p. 41).

Sindzingre, P., G. Ciccotti, C. Massobrio, and D. Frenkel (1987). “Partial enthalpies and related quantities in mixtures from computer simulation”. In: *Chemical Physics Letters* 36.1, p. 35 (cit. on p. 35).

Span, R. (2000). *Multiparameter Equations of State.* Berlin: Springer Verlag (cit. on p. 26).

Spoel D., van Maaren P. van der and H. Berendsen (1998). In: *The Journal of Chemical Physics* 108, p. 10220 (cit. on p. 22).

Steele, W.A. (1969). *Transport Phenomena in fluids.* New York: Marcel Dekker (cit. on p. 29).

Steinhauser, O. (1982). In: *Molecular Physics* 45, p. 335 (cit. on p. 22).

Stoll, J. (2004). “Molecular Models for the Prediction of Thermophysical Properties of Pure Fluids and Mixtures”. PhD thesis. Universität Stuttgart (cit. on pp. 39, 41, 42).

Stoll, J., J. Vrabec, and H. Hasse (2003). “A set of molecular models for carbon monoxide and halogenated hydrocarbons”. In: *The Journal of Chemical Physics* 119, p. 11396. DOI: `10.1063/1.1623475` (cit. on pp. 39–42).

Stone, A. J. (2008). “Intermolecular Potentials”. In: *Science* 321, p. 787 (cit. on p. 20).

The Economist (2010). March 13th (cit. on p. 1).

Vrabec, J. and H. Hasse (2002). In: *Molecular Physics* 100, p. 3375 (cit. on p. 35).

Vrabec, J., Y. Huang, and H. Hasse (2009). “Molecular models for 267 binary mixtures validated by vapor liquid quilibria: A systematic approach”. In: *Fluid Phase Equilibria* 279.2, pp. 120–135. DOI: `10.1016/j.fluid.2009.02.017` (cit. on p. 35).

Vrabec, J., J. Stoll, and H. Hasse (2001). “A Set of Molecular Models for Symmetric Quadrupolar Fluids”. In: *The Journal of Physical Chemistry B* 105.48, p. 12126. DOI: `10.1021/jp012542o` (cit. on pp. 39–42).

Widom, B. (1963). In: *The Journal of Chemical Physics* 39, p. 2808 (cit. on pp. 33, 35).

List of symbols and abbreviations

Symbols

symbol	meaning
A	Helmholtz energy
a	probability that determines whether the newly generated state is accepted or not
B	intermolecular potential
c_{p}	molar isobaric heat capacity
c_{v}	molar isochoric heat capacity
D	Maxwell-Stefan diffusion coefficient
E	total hamiltonian of the system
E	internal energy
F	molecular degrees of freedom in the system
G	Gibbs energy
g	radial distribution function
H	enthalpy
H	Henry's law constant
I	matrix of angular momentum of inertia
j_{e}	electric current flux
J_{p}	stress tensor
K	total kinetic energy of the system
k_{B}	Boltzmann's constant
l	length
L	phenomenological coefficients which are related to transport diffusion in a mixture
M	molar mass
m	mass
n_i	solvation number around molecule i
N_{A}	Avogadro constant
N	number of molecules
p	pressure
P	probability for a configuration
q	point charges magnitude
Q	quadrupole moment
R	molar or universal gas constant
r	distance
r_{c}	cut-off radius
S	entropy
T	temperature

symbol	**meaning**
t	time
U	total potential energy of the system
u	intermolecular potential energy
V	volume
v	velocity
w	speed of sound
w	angular velocity
x	composition of the liquid

Greek Symbols

symbol	**meaning**
α	trial probability of occurrence of a certain based on the current configuration
α_p	volume expansivity
β	inverse temperature
β_T	isothermal compressibility
ε	Lennard-Jones energy parameter
ϵ_0	permittivity of the vacuum
η	combination rule parameter
η	shear viscosity
η_v	bulk viscosity
Γ	torque
λ	thermal conductivity
μ	dipole moment
ϕ	azimuthal angle, to the positive x axis
φ	angle
π	transition probability between two states
ρ	number density
σ	Lennard-Jones size parameter
θ	polar angle, to the positive z axis
θ	unit step function
Θ	Heaviside function
$\tilde{\mu}$	chemical potential
ξ	combination rule parameter

Subscript/Superscript

symbol	**meaning**
$*$	reduced physical quantities
inf	infinite dilution

symbol	**meaning**
D	Dipole
id	ideal gas contribution
L	local
LJ	Lennard-Jones
q	Point charge
Q	Quadrupole
ref	reference
res	residual part
x	vector components
y	vector components
z	vector components

List of Abbreviations

H-bonding	Hydrogen bonding
NpH	Isoenthalpic–isobaric ensemble
LJ sites	Lennard-Jones sites
LRC	long range correction
MC	Monte-Carlo
MD	Molecular dynamics
NpT	isobaric-isothermal ensemble
NVE	micro-canonical ensemble
NVT	canonical ensemble
pseudo-μVT	grand equilibrium method
RDF	radial distribution function
RF	reaction field
VLE	Vapour-liquid equilibrium